

Math 316 - Exam #1

Name: Seth Recks (2 points will be lost for not writing your name)

Score: 61 / 65 = 93.8 %

1. Solve the following system of equations using the Gaussian elimination. You must show your work to get full credit. Please write your answer in the form $\{(x_1, x_2, x_3)\}$ in the space provided below.

$$x_1 + 2x_2 + x_3 = 4$$

$$3x_1 + 8x_2 + 7x_3 = 20$$

$$2x_1 + 7x_2 + 9x_3 = 23$$

$$\begin{bmatrix} 1 & 2 & 1 & 4 \\ 3 & 8 & 7 & 20 \\ 2 & 7 & 9 & 23 \end{bmatrix}$$

$$-3R_1 + R_2 \Rightarrow R_2$$

$$\begin{bmatrix} 1 & 2 & 1 & 4 \\ 0 & 2 & 4 & 8 \\ 2 & 7 & 9 & 23 \end{bmatrix}$$

$$-2R_1 + R_3 \Rightarrow R_3$$

$$\begin{bmatrix} 1 & 2 & 1 & 4 \\ 0 & 2 & 4 & 8 \\ 0 & 3 & 7 & 15 \end{bmatrix}$$

$$\frac{1}{2}R_2 \Rightarrow R_2$$

$$\begin{bmatrix} 1 & 2 & 1 & 4 \\ 0 & 1 & 2 & 4 \\ 0 & 3 & 7 & 15 \end{bmatrix}$$

$$-3R_2 + R_3 \Rightarrow R_3$$

$$\begin{bmatrix} 1 & 2 & 1 & 4 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$$-2R_2 + R_1 \Rightarrow R_1$$

$$\begin{bmatrix} 1 & 0 & -3 & -4 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$$-2R_3 + R_2 \Rightarrow R_2$$

$$\begin{bmatrix} 1 & 0 & -3 & -4 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$$3R_3 + R_1 \Rightarrow R_1$$

$$\begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$$\{(5, -2, 3)\}$$

Answer:

$$\{(5, -2, 3)\}$$

Points: (5) 3 0

2. Solve the following system of equations using Gaussian elimination. You must show your work to get full credit. Please write your answer in the form $\{(x_1, x_2, x_3, x_4)\}$ in the space provided below.

$$x_1 + x_2 + x_3 + x_4 = 12$$

$$x_1 + 2x_2 + 5x_4 = 17$$

$$3x_1 + 2x_2 + 4x_3 - x_4 = 31$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 12 \\ 1 & 2 & 0 & 5 & 17 \\ 3 & 2 & 4 & -1 & 31 \end{bmatrix}$$

$$-R_1 + R_2 \Rightarrow R_2$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 12 \\ 0 & 1 & -1 & 4 & 5 \\ 3 & 2 & 4 & -1 & 31 \end{bmatrix}$$

$$-3R_1 + R_3 \Rightarrow R_3$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 12 \\ 0 & 1 & -1 & 4 & 5 \\ 0 & -1 & 1 & -4 & -5 \end{bmatrix}$$

$$-R_2 + R_1 \Rightarrow R_1$$

$$\begin{bmatrix} 1 & 0 & 2 & -3 & 7 \\ 0 & 1 & -1 & 4 & 5 \\ 0 & -1 & 1 & -4 & -5 \end{bmatrix}$$

$$R_2 + R_3 \Rightarrow R_3$$

$$\begin{bmatrix} 1 & 0 & 2 & -3 & 7 \\ 0 & 1 & -1 & 4 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_1 + 2x_3 - 3x_4 = 7$$

$$x_2 - x_3 + 4x_4 = 5$$

$$x_3 = x_3$$

$$x_1 = -2x_3 + 3x_4 + 7$$

$$x_2 = x_3 - 4x_4 + 5$$

$$x_3 = x_3$$

$$x_4 = x_4$$

$$\text{if } x_3 = 1$$

and

$$x_4 = 0$$

$$x_1 = 5$$

$$x_2 = 6$$

$$5 + 6 + 1 + 0 = 12 \checkmark$$

$$5 + 2(6) + 0 = 17 \checkmark$$

$$3(5) + 2(6) + 4(1) + 0 = 31 \checkmark$$

$$\{(-2x_3 + 3x_4 + 7, x_3 - 4x_4 + 5, x_3, x_4)\}$$

Answer:

$$\{(-2x_3 + 3x_4 + 7, x_3 - 4x_4 + 5, x_3, x_4)\}$$

Points: 5 3 0

3. Given that $A = \begin{bmatrix} 4 & 1 & -2 & 7 \\ 3 & 1 & -1 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 5 \\ 3 & -1 \\ -2 & 4 \\ 2 & -3 \end{bmatrix}$, find the product AB . You must show your work to get full credit.

$$A \times B$$

$$2 \times \textcircled{4} \quad \textcircled{4} \times 2$$

$$\begin{bmatrix} 4 & 1 & -2 & 7 \\ 3 & 1 & -1 & 5 \end{bmatrix} \begin{bmatrix} 1 & 5 \\ 3 & -1 \\ -2 & 4 \\ 2 & -3 \end{bmatrix} = \begin{bmatrix} 25 & -10 \\ 18 & -5 \end{bmatrix}$$

$$(4)(1) + (1)(3) + (-2)(-2) + (7)(2) = 25$$

$$(4)(5) + (1)(-1) + (-2)(4) + (7)(-3) = -10$$

$$(3)(1) + (1)(3) + (-1)(-2) + (5)(2) = 18$$

$$(3)(5) + (1)(-1) + (-1)(4) + (5)(-3) = -5$$

Answer:

$$\begin{bmatrix} 25 & -10 \\ 18 & -5 \end{bmatrix}$$

Points: $\textcircled{5}$ 3 0

4. Solve the following homogeneous system of equations and write the answer as a linear combination. You must show your work to get full credit.

$$x_1 + 3x_2 - 15x_3 + 7x_4 = 0$$

$$x_1 + 4x_2 - 19x_3 + 10x_4 = 0$$

$$2x_1 + 5x_2 - 26x_3 + 11x_4 = 0$$

$$\begin{bmatrix} 1 & 3 & -15 & 7 \\ 1 & 4 & -19 & 10 \\ 2 & 5 & -26 & 11 \end{bmatrix}$$

$$-3R_2 + R_1 \Rightarrow R_1$$

$$\begin{bmatrix} 1 & 0 & -117 & -2 \\ 0 & 1 & 34 & 3 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$-R_1 + R_2 \Rightarrow R_2$$

$$\begin{bmatrix} 1 & 3 & -15 & 7 \\ 0 & 1 & 34 & 3 \\ 2 & 5 & -26 & 11 \end{bmatrix}$$

$$-34R_3 + R_2 \Rightarrow R_2$$

$$\begin{bmatrix} 1 & 0 & -117 & -2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$-2R_1 + R_3 \Rightarrow R_3$$

$$\begin{bmatrix} 1 & 3 & -15 & 7 \\ 0 & 1 & 34 & 3 \\ 0 & -1 & 4 & -3 \end{bmatrix}$$

$$117R_3 + R_1 \Rightarrow R_1$$

$$\begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$x_1 = 2x_4$$

$$x_2 = -3x_4$$

$$x_3 = 0$$

$$x_4 = x_4$$

$$x_4 = 2 \dots$$

$$x_1 = 4$$

$$x_2 = -6$$

$$x_3 = 0$$

$$x_4 = 2$$

$$4 + 3(-6) + 7(2) = 0 \checkmark$$

$$4 + 4(-6) + 10(2) = 0 \checkmark$$

$$2(4) + 5(-6) + 11(2) = 0 \checkmark$$

$$R_2 + R_3 \Rightarrow R_3$$

$$\begin{bmatrix} 1 & 3 & -15 & 7 \\ 0 & 1 & 34 & 3 \\ 0 & 0 & 38 & 0 \end{bmatrix}$$

$$\frac{1}{38}R_3 \Rightarrow R_3$$

$$\begin{bmatrix} 1 & 3 & -15 & 7 \\ 0 & 1 & 34 & 3 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$X = x_4 \begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \end{bmatrix}$$

linear combination form:

$$2x_4 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + -3x_4 \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} + 0 \begin{bmatrix} -15 \\ -19 \\ -26 \end{bmatrix} + x_4 \begin{bmatrix} 7 \\ 10 \\ 11 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Answer:

$$2x_4 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + -3x_4 \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} + 0 \begin{bmatrix} -15 \\ -19 \\ -26 \end{bmatrix} + x_4 \begin{bmatrix} 7 \\ 10 \\ 11 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Points: 5 (3) 0

5. Find all solutions to $Ax = 0$ for the matrix $A = \begin{bmatrix} 3 & -4 & 2 \\ -9 & 12 & -6 \\ -6 & 8 & -4 \end{bmatrix}$. Please write your answer in the form $\{(x_1, x_2, x_3)\}$ in the space provided below.

$$\begin{bmatrix} 3 & -4 & 2 \\ -9 & 12 & -6 \\ 6 & 8 & -4 \end{bmatrix}$$

$$\frac{1}{3}R_1 \Rightarrow R_1$$

$$\begin{bmatrix} 1 & -\frac{4}{3} & \frac{2}{3} \\ -9 & 12 & -6 \\ 6 & 8 & -4 \end{bmatrix}$$

$$-6R_1 + R_3 \Rightarrow R_3$$

$$\begin{bmatrix} 1 & -\frac{4}{3} & \frac{2}{3} \\ -9 & 12 & -6 \\ 0 & 16 & -8 \end{bmatrix}$$

$$9R_1 + R_2 \Rightarrow R_2$$

$$\begin{bmatrix} 1 & -\frac{4}{3} & \frac{2}{3} \\ 0 & 0 & 0 \\ 0 & 16 & -8 \end{bmatrix}$$

$$x_1 = \frac{4}{3}x_2 - \frac{2}{3}x_3 \rightarrow \frac{4}{3}x_2 - \frac{2}{3}(2x_2) = \frac{4}{3}x_2 - \frac{4}{3}x_2 = 0$$

$$x_2 = x_2$$

$$-8x_3 = -16x_2 \rightarrow x_3 = 2x_2$$

$$\{(0, x_2, 2x_2)\}$$

$$3x_1 - 4x_2 + 2x_3 = 0$$

$$-9x_1 + 12x_2 - 6x_3 = 0$$

$$-6x_1 + 8x_2 - 4x_3 = 0$$

$$\text{pf } x_2 = 2 \dots$$

$$-4(2) + 2(2)(2) = 0 \checkmark$$

$$12(2) - 6(2)(2) = 0 \checkmark$$

$$8(2) - 4(2)(2) = 0 \checkmark$$

Answer:

$$\{(0, x_2, 2x_2)\}$$

Points: 5 3 0

6. Solve the nonhomogeneous equation $Ax = b$, given $A = \begin{bmatrix} -5 & 8 \\ 10 & -16 \end{bmatrix}$ and

$b = \begin{bmatrix} -21 \\ 42 \end{bmatrix}$. Express your solution x in the form $x = x_p + x_h$, where x_p is the particular solution and x_h is the solution to the homogeneous equation $Ax = 0$. You must show your work to get full credit.

$$\left[\begin{array}{cc|c} -5 & 8 & -21 \\ 10 & -16 & 42 \end{array} \right]$$

$$-1/5 R_1 \Rightarrow R_1$$

$$\left[\begin{array}{cc|c} 1 & -8/5 & 21/5 \\ 10 & -16 & 42 \end{array} \right]$$

$$-10 R_1 + R_2 \Rightarrow R_2$$

$$\left[\begin{array}{cc|c} 1 & -8/5 & 21/5 \\ 0 & 0 & 0 \end{array} \right]$$

$$x_1 + -8/5 x_2 = 21/5$$

$$x_1 = 8/5 x_2 + 21/5$$

$$x_2 = x_2 + 0$$

$$x_p = \begin{bmatrix} 21/5 \\ 0 \end{bmatrix}$$

$$x_h = x_2 \begin{bmatrix} 8/5 \\ 1 \end{bmatrix}$$

$$x = x_p + x_h = \begin{bmatrix} 21/5 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 8/5 \\ 1 \end{bmatrix}$$

Answer:

$$x = \begin{bmatrix} 21/5 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 8/5 \\ 1 \end{bmatrix}$$

Points: 5 3 0

7. Do the vectors $v_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$, $v_2 = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$ and $v_3 = \begin{bmatrix} 3 \\ 8 \\ 7 \end{bmatrix}$ span $\mathcal{R}^3$? Only answer yes or no. You must justify your answer by showing your work to get full credit.

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 8 \\ 2 & 4 & 7 \end{bmatrix}$$

$$-2R_1 + R_2 \Rightarrow R_2$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 2 \\ 2 & 4 & 7 \end{bmatrix}$$

$$-2R_1 + R_3 \Rightarrow R_3$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$-R_2 \Rightarrow R_2$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$-2R_2 + R_1 \Rightarrow R_1$$

$$\begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$2R_3 + R_2 \Rightarrow R_2$$

$$\begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$-7R_3 + R_1 \Rightarrow R_1$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Yes.

Answer:

Yes.

Points: 5 3 0

8. Determine if the vectors $v_1 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$, $v_2 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$ and $v_3 = \begin{bmatrix} 1 \\ 5 \\ 3 \end{bmatrix}$ in $\mathcal{R}^3$ are linearly independent. Only answer yes or no. You must show your work to get full credit.

$$\begin{bmatrix} -1 & 3 & 1 \\ 2 & 1 & 5 \\ 1 & 1 & 3 \end{bmatrix}$$

Row of all zeros. No.

$$-R_1 \Rightarrow R_1$$

$$\begin{bmatrix} 1 & -3 & -1 \\ 2 & 1 & 5 \\ 1 & 1 & 3 \end{bmatrix}$$

$$-2R_1 + R_2 \Rightarrow R_2$$

$$\begin{bmatrix} 1 & -3 & -1 \\ 0 & 7 & 7 \\ 1 & 1 & 3 \end{bmatrix}$$

$$-R_1 + R_3 \Rightarrow R_3$$

$$\begin{bmatrix} 1 & -3 & -1 \\ 0 & 7 & 7 \\ 0 & 4 & 4 \end{bmatrix}$$

$$\frac{1}{7}R_2 \Rightarrow R_2$$

$$\begin{bmatrix} 1 & -3 & -1 \\ 0 & 1 & 1 \\ 0 & 4 & 4 \end{bmatrix}$$

$$-4R_2 + R_3 \Rightarrow R_3$$

$$\begin{bmatrix} 1 & -3 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Answer:

No.

Points: 5 3 0

9. Find the inverse of the matrix $A = \begin{bmatrix} 4 & 3 & 2 \\ 5 & 6 & 3 \\ 3 & 5 & 2 \end{bmatrix}$. You must show your work in order to get full credit.

$$\left[\begin{array}{ccc|ccc} 4 & 3 & 2 & 1 & 0 & 0 \\ 5 & 6 & 3 & 0 & 1 & 0 \\ 3 & 5 & 2 & 0 & 0 & 1 \end{array} \right]$$

$$\frac{1}{4}R_1 \Rightarrow R_1$$

$$\left[\begin{array}{ccc|ccc} 1 & \frac{3}{4} & \frac{1}{2} & \frac{1}{4} & 0 & 0 \\ 5 & 6 & 3 & 0 & 1 & 0 \\ 3 & 5 & 2 & 0 & 0 & 1 \end{array} \right]$$

$$-5R_1 + R_2 \Rightarrow R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & \frac{3}{4} & \frac{1}{2} & \frac{1}{4} & 0 & 0 \\ 0 & \frac{9}{4} & \frac{1}{2} & -\frac{5}{4} & 1 & 0 \\ 3 & 5 & 2 & 0 & 0 & 1 \end{array} \right]$$

$$-3R_1 + R_3 \Rightarrow R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & \frac{3}{4} & \frac{1}{2} & \frac{1}{4} & 0 & 0 \\ 0 & \frac{9}{4} & \frac{1}{2} & -\frac{5}{4} & 1 & 0 \\ 0 & \frac{5}{4} & \frac{1}{2} & -\frac{3}{4} & 0 & 1 \end{array} \right]$$

$$\frac{4}{9}R_2 \Rightarrow R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & \frac{3}{4} & \frac{1}{2} & \frac{1}{4} & 0 & 0 \\ 0 & 1 & \frac{2}{9} & -\frac{5}{9} & \frac{4}{9} & 0 \\ 0 & \frac{5}{4} & \frac{1}{2} & -\frac{3}{4} & 0 & 1 \end{array} \right]$$

$$-\frac{1}{4}R_2 + R_3 \Rightarrow R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & \frac{3}{4} & \frac{1}{2} & \frac{1}{4} & 0 & 0 \\ 0 & 1 & \frac{2}{9} & -\frac{5}{9} & \frac{4}{9} & 0 \\ 0 & 0 & -\frac{1}{9} & \frac{7}{9} & -\frac{1}{9} & 1 \end{array} \right]$$

$$-\frac{3}{4}R_2 + R_1 \Rightarrow R_1$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & \frac{1}{3} & \frac{4}{3} & -\frac{1}{3} & 0 \\ 0 & 1 & \frac{2}{9} & -\frac{5}{9} & \frac{4}{9} & 0 \\ 0 & 0 & -\frac{1}{9} & \frac{7}{9} & -\frac{1}{9} & 1 \end{array} \right]$$

$$-9R_3 \Rightarrow R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & \frac{1}{3} & \frac{4}{3} & -\frac{1}{3} & 0 \\ 0 & 1 & \frac{2}{9} & -\frac{5}{9} & \frac{4}{9} & 0 \\ 0 & 0 & 1 & -7 & 11 & -9 \end{array} \right]$$

$$-\frac{2}{9}R_3 + R_2 \Rightarrow R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & \frac{1}{3} & \frac{4}{3} & -\frac{1}{3} & 0 \\ 0 & 1 & 0 & 1 & -2 & 2 \\ 0 & 0 & 1 & -7 & 11 & -9 \end{array} \right]$$

$$-\frac{1}{3}R_3 + R_1 \Rightarrow R_1$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -4 & 3 \\ 0 & 1 & 0 & 1 & -2 & 2 \\ 0 & 0 & 1 & -7 & 11 & -9 \end{array} \right]$$

Answer:

$$\begin{bmatrix} 3 & -4 & 3 \\ 1 & -2 & 2 \\ -7 & 11 & -9 \end{bmatrix}$$

Points: 5 3 0

10. Find $\det \begin{bmatrix} 5 & -2 & -3 \\ 4 & 0 & 1 \\ 3 & -1 & 2 \end{bmatrix}$. You must show your work in order to get full credit.

$$\begin{array}{rrr} 5 & -2 & -3 \\ 4 & 0 & 1 \\ 3 & -1 & 2 \end{array}$$

$$(5)(0)(2) + (-2)(1)(3) + (-3)(4)(-1) - (-2)(4)(2) - (5)(1)(-1) - (-3)(0)(3)$$
$$0 + -6 + 12 + 16 + 5 + 0 = 27$$

Answer:

27

Points: 5 3 0

11. Fill in the missing parts, A, B, and C of the Invertible Matrix Theorem in the space provided below

Let A be a ____ A ____ matrix. The following statements are equivalent:

1. The equation $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b} \in \mathbb{R}^n$.
2. Every vector $\mathbf{b} \in \mathbb{R}^n$ is a linear combination of the columns of A.
3. ____ B ____.
4. A has a pivot position in every row and column. That is, when the matrix A is row reduced, there are no rows of all zeroes.
5. The columns of A are linearly independent
6. ____ C ____.
7. A is invertible (has an inverse matrix)
8. The columns of A form a basis for $\mathbb{R}^n$.

Answer:

A - $n \times n$

B - The span of the columns of A is $\mathbb{R}^n$.

C - $\det(A) \neq 0$

Points: 5 3 0

12. Is the plane $2x + 4y + 3z = 0$ a subspace of $\mathbb{R}^3$. You must show your work in order to get full credit. Please answer yes or no in the space provided.

Closed under addition?

$$2x_1 + 4y_1 + 3z_1 = 0$$

$$2x_2 + 4y_2 + 3z_2 = 0$$

$$+ \begin{array}{l} 2x_1 + 4y_1 + 3z_1 = 0 \\ 2x_2 + 4y_2 + 3z_2 = 0 \end{array}$$

$$2(x_1 + x_2) + 4(y_1 + y_2) + 3(z_1 + z_2) = 0 + 0 = 0 \quad \text{yes} \checkmark$$

Closed under multiplication?

$$c \in \mathbb{R}$$

$$c(2x_1 + 4y_1 + 3z_1 = 0)$$

$$2(cx_1) + 4(cy_1) + 3(cz_1) = c(0) = 0 \quad \text{yes} \checkmark$$

Contains zero element?

$$z = (0, 0, 0)$$

$$2(0) + 4(0) + 3(0) = 0 \quad \text{yes} \checkmark$$

Yes.

Answer:

Yes.

Points: 5 3 0

13. Let $v_1 = \begin{bmatrix} 1 \\ -1 \\ -2 \\ -3 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 3 \end{bmatrix}$, $v_3 = \begin{bmatrix} 1 \\ -1 \\ -3 \\ -2 \end{bmatrix}$ and $v_4 = \begin{bmatrix} 0 \\ 3 \\ -1 \\ 2 \end{bmatrix}$. Determine if these vectors

form a basis for $\mathbb{R}^4$? Please write yes or no and be sure to show all of your work in order to get full credit.

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ -1 & -1 & -1 & 3 \\ -2 & 2 & -3 & -1 \\ -3 & 3 & -2 & 2 \end{bmatrix}$$

$R_2 \leftrightarrow R_4$ (to avoid 0 in pivot position #2)

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ -3 & 3 & -2 & 2 \\ -2 & 2 & -3 & -1 \\ -1 & -1 & -1 & 3 \end{bmatrix}$$

$3R_1 + R_2 \Rightarrow R_2$

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 6 & 1 & 2 \\ -2 & 2 & -3 & -1 \\ -1 & -1 & -1 & 3 \end{bmatrix}$$

$2R_1 + R_3 \Rightarrow R_3$

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 6 & 1 & 2 \\ 0 & 4 & -1 & -1 \\ -1 & -1 & -1 & 3 \end{bmatrix}$$

$R_1 + R_4 \Rightarrow R_4$

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 6 & 1 & 2 \\ 0 & 4 & -1 & -1 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

$\frac{1}{6}R_2 \Rightarrow R_2$

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & \frac{1}{6} & \frac{1}{3} \\ 0 & 4 & -1 & -1 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

$-4R_2 + R_3 \Rightarrow R_3$

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & \frac{1}{6} & \frac{1}{3} \\ 0 & 0 & -\frac{5}{3} & -\frac{7}{3} \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

$-R_2 + R_1 \Rightarrow R_1$

$$\begin{bmatrix} 1 & 0 & \frac{5}{6} & -\frac{1}{3} \\ 0 & 1 & \frac{1}{6} & \frac{1}{3} \\ 0 & 0 & -\frac{5}{3} & -\frac{7}{3} \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

$-\frac{3}{5}R_3 \Rightarrow R_3$

$$\begin{bmatrix} 1 & 0 & \frac{5}{6} & -\frac{1}{3} \\ 0 & 1 & \frac{1}{6} & \frac{1}{3} \\ 0 & 0 & 1 & \frac{7}{5} \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

$-\frac{1}{6}R_3 + R_2 \Rightarrow R_2$

$$\begin{bmatrix} 1 & 0 & \frac{5}{6} & -\frac{1}{3} \\ 0 & 1 & 0 & \frac{1}{10} \\ 0 & 0 & 1 & \frac{7}{5} \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

$-\frac{5}{6}R_3 + R_1 \Rightarrow R_1$

$$\begin{bmatrix} 1 & 0 & 0 & -\frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{10} \\ 0 & 0 & 1 & \frac{7}{5} \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

$\frac{1}{3}R_4 \Rightarrow R_4$

$$\begin{bmatrix} 1 & 0 & 0 & -\frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{10} \\ 0 & 0 & 1 & \frac{7}{5} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$-\frac{7}{5}R_4 + R_3 \Rightarrow R_3$

$-\frac{1}{10}R_4 + R_2 \Rightarrow R_2$

$\frac{3}{2}R_4 + R_1 \Rightarrow R_1$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Yes!

Vectors v_1, v_2, v_3 and v_4 form a basis for $\mathbb{R}^4$.

Answer:

Yes.

Points: 5 3 0

Math 316 - Exam #2

Name: Seth Ricks (2 points will be lost for not writing your name)

Score: 64 / 70 = 91.4 %
66 94.3

1. Classify the following differential equation by order

a. $y'' + y' = 3x \sin y$

Second order differential equation

Answer:

Second Order

Points: 5 0

b. $3x^3 y'''' - x^3 y' + e^{xy} y = 0$

Fourth order differential equation

Answer:

~~Fourth~~ order

Points: 5 3 0

2. Solve the differential equation $y' = x\sqrt{x^2 + 9}$, $y(-4) = 0$

$$y' = x\sqrt{x^2 + 9}$$

$$y = \int x\sqrt{x^2 + 9} \, dx$$

$$u = x^2 + 9$$

$$\frac{du}{dx} = 2x$$

$$dx = \frac{du}{2x}$$

$$y = \int x\sqrt{u} \frac{du}{2x} = \frac{1}{2} \int \sqrt{u} \, du$$

$$= \frac{1}{2} \cdot \frac{2}{3} u^{3/2} + C = \frac{1}{3} u^{3/2} + C$$

$$y = \frac{1}{3} (x^2 + 9)^{3/2} + C$$

Check:

$$y' = \frac{1}{3} \cdot \frac{3}{2} (x^2 + 9)^{1/2} \cdot 2x + 0$$

$$= x\sqrt{x^2 + 9} \quad \checkmark$$

$$y(-4) = 0 = \frac{1}{3} ((-4)^2 + 9)^{3/2} + C$$

$$= \frac{125}{3} + C$$

$$C = -\frac{125}{3}$$

$$y = \frac{1}{3} (x^2 + 9)^{3/2} - \frac{125}{3}$$

Answer:

$$y = \frac{1}{3} (x^2 + 9)^{3/2} - \frac{125}{3}$$

Points: 5 3 0

3. Solve the differential equation $y' - 3t^4y = 0$.

$$y' - 3t^4y = 0$$

$$p(t) = \int -3t^4 dt = -3 \int t^4 = -3 \cdot \frac{1}{5} t^5 = -\frac{3}{5} t^5$$

$$e^{p(t)} = e^{-\frac{3}{5} t^5}$$

$$\left[y e^{-\frac{3}{5} t^5} \right]' = 0$$

$$y e^{-\frac{3}{5} t^5} = \int 0 = C$$

$$y = C e^{\frac{3}{5} t^5}$$

Check:

$$y = C e^{\frac{3}{5} t^5}$$

$$y' = C e^{\frac{3}{5} t^5} \cdot 3t^4$$

$$= C 3t^4 e^{\frac{3}{5} t^5}$$

$$C 3t^4 e^{\frac{3}{5} t^5} - 3t^4 (C e^{\frac{3}{5} t^5}) = 0$$

$$C 3t^4 e^{\frac{3}{5} t^5} - C 3t^4 e^{\frac{3}{5} t^5} = 0 \checkmark$$

Answer:

$$y = C e^{\frac{3}{5} t^5}$$

Points: (5) 3 0

4. Solve the differential equation $y' - \frac{2}{t}y = \frac{2}{3}t^4$.

$$P(t) = \int -\frac{2}{t} dt = -2 \int \frac{1}{t} dt = -2 \ln|t| = \ln t^{-2}$$

$$e^{P(t)} = e^{\ln t^{-2}} = t^{-2}$$

$$[y t^{-2}]' = \frac{2}{3} t^4 \cdot t^{-2} = \frac{2}{3} t^2$$

$$y t^{-2} = \int \frac{2}{3} t^2 dt$$

$$= \frac{2}{3} \int t^2 dt = \frac{2}{3} \cdot \frac{1}{3} t^3 + C = \frac{2}{9} t^3 + C$$

$$y = t^2 \left(\frac{2}{9} t^3 + C \right) = \frac{2}{9} t^5 + C t^2$$

$$y = \frac{2}{9} t^5 + C t^2$$

Check:

$$y = \frac{2}{9} t^5 + C t^2$$

$$y' = \frac{10}{9} t^4 + 2Ct$$

$$\frac{10}{9} t^4 + 2Ct - \frac{2}{t} \left[\frac{2}{9} t^5 + C t^2 \right] = \frac{2}{3} t^4$$

$$\frac{10}{9} t^4 + 2Ct - \frac{4}{9} t^4 - 2Ct = \frac{2}{3} t^4$$

$$\left(\frac{10}{9} - \frac{4}{9} \right) t^4 = \frac{2}{3} t^4$$

$$\frac{6}{9} t^4 = \frac{2}{3} t^4$$

$$\frac{2}{3} t^4 = \frac{2}{3} t^4 \quad \checkmark$$

Answer:

$$y = \frac{2}{9} t^5 + C t^2$$

Points: 5 3 0

5. A ball is dropped from a height of 100 feet with zero initial velocity. Assuming no air resistance, what is the time required to reach the ground? Round answer to 1 decimal point to the right of the decimal.

$$a(t) = g$$

$$x_0 = 100$$

$$v(t) = gt + v_0$$

$$v_0 = 0$$

$$x(t) = \frac{1}{2}gt^2 + v_0t + x_0$$

$$x(t) = \frac{1}{2}(-32)t^2 + 0t + 100$$

$$0 = -16t^2 + 100$$

$$0 = -16\left(t^2 - \frac{100}{16}\right)$$

$$0 = -16\left(t^2 - \frac{25}{4}\right)$$

$$t^2 = \frac{25}{4}$$

$$t = \sqrt{\frac{25}{4}} = \frac{5}{2} = 2.5 \text{ Seconds}$$

Answer:

2.5

Points: 5 3 0

Not needed because
we aren't solving for A

6. A tank initially holds 80 gallons of a brine solution containing $\frac{1}{8}$ pounds of salt per gallon.

At $t = 0$, another brine solution containing 1 pound of salt per gallon is poured into the tank at the rate of $4 \frac{\text{gal}}{\text{min}}$, while the well-stirred mixture leaves the tank at the rate of $8 \frac{\text{gal}}{\text{min}}$.

Set up the differential equation and initial conditions for this mixing problem. YOU DO NOT NEED TO SOLVE THE DIFFERENTIAL EQUATION!!

$$V_0 = 80 \quad V = 80 + 4t - 8t = 80 - 4t$$

$$r_i = 4 \quad r_o = 8$$

$$C_i = 1 \quad C_o = \frac{A}{V} = \frac{A}{80 - 4t}$$

$$f_i = 4 \cdot 1 = 4 \quad f_o = 8 \cdot \frac{A}{80 - 4t} = \frac{8A}{80 - 4t} = \frac{2A}{20 - t}$$

$$A' = f_i - f_o = 4 - \frac{2A}{20 - t}$$

$$A' = 4 - \frac{2A}{20 - t}$$

Answer:

$$A' = 4 - \frac{2A}{20 - t} \quad A(0) = ?$$

Points: 5 (3) 0

7. A certain radioactive material is known to decay at a rate proportional to the amount present. If initially there is 50 milligrams of the material present and after two hours it is observed that the material has lost 10 percent of its original mass, find the time at which the material has decayed to one half of its initial mass. Round answer to nearest hour.

$$P = P_0 e^{kt}$$

$$10\% \text{ of } 50 = 5$$

$$P_0 = 50$$

$$\frac{1}{2} \text{ of } 50 = 25$$

$$P(2) = 45$$

$$P(?) = 25$$

$$P(t) = 50 e^{kt}$$

$$P(2) = 45 = 50 e^{2k}$$

$$\frac{45}{50} = \frac{9}{10} = e^{2k}$$

$$\frac{\ln \frac{9}{10}}{2} = k$$

$$P(t) = 50 e^{\frac{\ln \frac{9}{10}}{2} t}$$

$$25 = 50 e^{\frac{\ln \frac{9}{10}}{2} t}$$

$$\ln \frac{1}{2} = \frac{\ln \frac{9}{10}}{2} t$$

$$t = \frac{2 \ln \frac{1}{2}}{\ln \frac{9}{10}} \approx 13 \text{ hours}$$

Answer:

13

Points: 5 3 0

8. Solve the differential equation $y' = \frac{x+1}{y^4+1}$ by the method of separation of variables. Write your answer in implicit form (i.e., do not try to solve for y).

$$y' = \frac{x+1}{y^4+1}$$

$$\frac{dy}{dx} = \frac{x+1}{y^4+1}$$

$$(y^4+1)dy = (x+1)dx$$

$$\int y^4+1 dy = \int (x+1) dx$$

$$\frac{1}{5}y^5+y = \frac{1}{2}x^2+x+C$$

$$\frac{1}{5}y^5+y - \frac{1}{2}x^2 - x + C = 0$$

Answer:

$$\frac{1}{5}y^5+y - \frac{1}{2}x^2 - x + C = 0$$

Points: (5) 3 0

9. Solve the differential equation $(x + \sin y)dx + (x \cos y - 2y)dy = 0$ using the exact method. Write your answer in implicit form (i.e., do not try to solve for y).

$$\underbrace{(x + \sin y)}_{M(x,y)} dx + \underbrace{(x \cos y - 2y)}_{N(x,y)} dy = 0$$

$$\frac{dM}{dy} = 0 + \cos y \quad \frac{dN}{dx} = \cos y - 0$$

$$\frac{dM}{dy} - \frac{dN}{dx} = 0 \quad ? \quad \checkmark$$

$$\int (x + \sin y) dx + \int (x \cos y - 2y) dy = 0$$

$$\frac{1}{2}x^2 + x \sin y + x \sin y - y^2 + C$$

$$\frac{1}{2}x^2 + x \sin y - y^2 + C = 0$$

Answer:

$$\frac{1}{2}x^2 + x \sin y - y^2 + C = 0$$

10. Consider an animal population $P(t)$ that is modeled by the equation

Find $P(t)$. $75 \frac{dP}{dt} = 0.05P(75 - P), P(0) = 10$ $P(t) = \frac{A}{1 + \left(\frac{A - P_0}{P_0}\right)e^{-Kt}}$

Correct form: $A \frac{dP}{dt} = KP(A - P)$

$$75 \frac{dP}{dt} = (75)(0.05)P(75 - P)$$

$$75 \frac{dP}{dt} = 3.75P(75 - P)$$

$$P(t) = \frac{75}{1 + \left(\frac{75 - 10}{10}\right)e^{-3.75t}}$$

$$P(t) = \frac{75}{1 + \left(\frac{13}{2}\right)e^{-3.75t}}$$

Answer:

$$P(t) = \frac{75}{1 + \left(\frac{13}{2}\right)e^{-3.75t}}$$

Points: (5) (8) 0

11. A metal bar at a temperature of 100°F is placed in a room at a constant temperature of 0°F . If after 20 minutes the temperature of the bar is 50°F , find the temperature of the bar after 10 minutes. Round your answer to two decimal places to the right of the decimal point.

$$T(t) = (T_0 - T_m)e^{-kt} + T_m$$

$$T_0 = 100$$

$$T_m = 0$$

$$T(20) = 50$$

$$T(10) = ?$$

$$T(t) = (100 - 0)e^{-kt} + 0$$

$$= 100e^{-kt}$$

$$T(20) = 50 = 100e^{-20k}$$

$$-\frac{\ln(\frac{1}{2})}{20} = k$$

$$T(t) = 100e^{\frac{\ln \frac{1}{2}}{20} t}$$

$$T(10) = 100e^{\frac{\ln \frac{1}{2}}{20} \cdot 10} \approx 70.71^{\circ}\text{F}$$

Answer:

70.71

Points: (5) 3 0

12. The cylindrical tank with a radius of 0.20 m contains water to a depth of 1.44 m of water. A bottom plug of the tank is pulled at time $t = 0$. The radius of the hole is 0.15 m. How long will it take before the tank is empty? Round your answer to four decimal places to the right of the decimal point.

$$h = \left(\sqrt{h_0} - \frac{r^2}{2R^2} \sqrt{2g} t \right)^2$$

$$h_0 = 1.44 \quad \sqrt{h_0} = 1.2$$

$$r = 0.15 \quad r^2 = 0.0225$$

$$R = 0.20 \quad 2R^2 = 0.08$$

$$h = 0$$

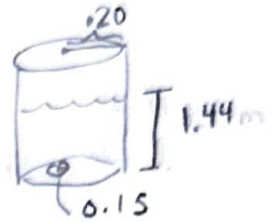
$$0 = \left(\sqrt{h_0} - \frac{r^2}{2R^2} \sqrt{2g} t \right)^2$$

$$-\sqrt{h_0} \cdot -\frac{2R^2}{r^2 \sqrt{2g}} = t$$

$$t = \frac{2R^2 \sqrt{h_0}}{r^2 \sqrt{2g}} = \frac{2(0.08)(1.2)}{(0.0225)\sqrt{19.6}} \approx 0.9637$$

Check:

$$0 = \left(\sqrt{1.2} - \frac{(0.15)^2}{2(0.20)^2} \sqrt{2 \cdot 9.8} \cdot 0.9637 \right)^2 = 0 \quad \checkmark$$



Answer:

0.9637

Points: 5 3 0

13. Find the approximate solution to the IVP $y' + y^3 = t^3$, $y(0) = 2$ at $y(1)$ using Euler's method. Use $h = 0.1$. Write your answer with 5 decimal digits to the right of the decimal point.

$$y' + y^3 = t^3; y(0) = 2; y(1) \approx ?$$

$$y' = t^3 - y^3$$

t_n	y_n
0	2
0.1	1.2
0.2	1.0273
0.3	0.919684378
0.4	0.844595693
0.5	0.790747145
0.6	0.753803224
0.7	0.73257067
0.8	0.727556548
0.9	0.740244145
1.0	0.772581623

$$y_6 = y_5 + (0.1)[y'(0.5, y_5)]$$

$$= 0.790747145 + (0.1)(-0.369439202)$$

$$= 0.753803224$$

$$y'(0.6, y_6) = 0.6^3 - (0.753803224)^3$$

$$= -0.212325541$$

$$y_7 = y_6 + (0.1)[y'(0.6, y_6)]$$

$$= (0.753803224) + (0.1)(-0.212325541)$$

$$= 0.73257067$$

$$y'(0.7, y_7) = 0.7^3 - (0.73257067)^3$$

$$= -0.056141219$$

$$y_8 = y_7 + (0.1)[y'(0.7, y_7)]$$

$$= 0.73257067 + (0.1)(-0.056141219)$$

$$= 0.727556548$$

$$y'(0.2) = 0^3 - 2^3 = -8$$

$$y_1 = y_0 + [y'(0.2)](0.1) = 2 + (0.1)(-8) = 1.2$$

$$y'(0.1, 1.2) = 0.1^3 - 1.2^3 = -1.727$$

$$y_2 = y_1 + (0.1)[y'(0.1, 1.2)] = 1.2 + (0.1)(-1.727) = 1.0273$$

$$y'(0.2, 1.0273) = 0.2^3 - (1.0273)^3 = -1.076156216$$

$$y_3 = y_2 + (0.1)[y'(0.2, 1.0273)] = 0.919684378$$

$$y'(0.3, y_3) = 0.3^3 - (0.919684378)^3 = -0.750886848$$

$$y_4 = y_3 + (0.1)[y'(0.3, y_3)] = 0.844595693$$

$$y'(0.4, y_4) = 0.4^3 - (0.844595693)^3 = -0.538485483$$

$$y_5 = y_4 + (0.1)[y'(0.4, y_4)] = 0.790747145$$

$$y'(0.5, y_5) = 0.5^3 - (0.790747145)^3 = -0.369439202$$

$$y'(0.8, y_8) = 0.8^3 - 0.727556548^3 = 0.126875968$$

$$y_9 = y_8 + (0.1)[y'(0.8, y_8)]$$

$$= 0.727556548 + (0.1)(0.126875968)$$

$$= 0.740244145$$

$$y'(0.9, y_9) = 0.9^3 - 0.740244145^3 = 0.323374787$$

$$y_{10} = y_9 + (0.1)[y'(0.9, y_9)]$$

$$= 0.740244145 + (0.1)(0.323374787)$$

$$= 0.772581623$$

$$y_{10} = y(1) \approx 0.77258$$

Answer:

0.77258

Points: 5 3 0

Math 316 - Exam #3

Name: Seth Rocks

(2 points will be lost for not writing your name)

Score: 45 / 45 = 100 %

1. Solve the differential equation $y'' - 36y' = 0$. You must show all of your work in order to get full credit.

$$r^2 - 36r = 0$$

$$r(r - 36) = 0$$

$$r = 0; r = 36$$

$$y = C_1 e^{0t} + C_2 e^{36t}$$

$$y = C_1 + C_2 e^{36t}$$

Check:

$$y = C_1 + C_2 e^{36t}$$

$$y' = 36C_2 e^{36t}$$

$$y'' = 1296C_2 e^{36t}$$

$$y'' - 36y' = 0?$$

$$1296C_2 e^{36t} - 36(36C_2 e^{36t}) = 0$$

$$1296C_2 e^{36t} - 1296C_2 e^{36t} = 0 \checkmark$$

Awesome!

Answer:

$$y = C_1 + C_2 e^{36t}$$

Points: 5 3 0

2. Solve the differential equation: $y'' - 18y' + 81y = 0$. You must show all of your work in order to get full credit.

$$r^2 - 18r + 81 = 0$$

$$(r-9)^2 = 0$$

$$r=9; r=9$$

$$y = C_1 e^{9t} + C_2 t e^{9t}$$

Check:

$$y = C_1 e^{9t} + C_2 t e^{9t}$$

$$y' = 9C_1 e^{9t} + C_2 t e^{9t} + C_2 e^{9t}$$

$$y'' = 81C_1 e^{9t} + 81C_2 t e^{9t} + 9C_2 e^{9t} + 9C_2 e^{9t}$$
$$= 81C_1 e^{9t} + 81C_2 t e^{9t} + 18C_2 e^{9t}$$

$$y'' - 18y' + 81y = 0?$$

$$\underline{81C_1 e^{9t}} + \underline{81C_2 t e^{9t}} + \underline{18C_2 e^{9t}} - \underline{162C_1 e^{9t}} - \underline{162C_2 t e^{9t}} - \underline{18C_2 e^{9t}} + \underline{81C_1 e^{9t}} + \underline{81C_2 t e^{9t}} = 0?$$

$0=0 \checkmark$

Answer:

$$y = C_1 e^{9t} + C_2 t e^{9t}$$

Points: 5 3 0

3. Solve the differential equation: $y'' + 2y' + 9y = 0$. You must show all of your work in order to get full credit.

$$r^2 + 2r + 9 = 0$$

$$r = \frac{-2 \pm \sqrt{2^2 - 4(1)(9)}}{2(1)} = \frac{-2 \pm 4\sqrt{2}i}{2} = -1 \pm 2\sqrt{2}i$$

$$y = C_1 e^{-t} \cos 2\sqrt{2}t + C_2 e^{-t} \sin 2\sqrt{2}t$$

Check:

$$y = C_1 e^{-t} \cos 2\sqrt{2}t + C_2 e^{-t} \sin 2\sqrt{2}t$$

$$y' = -C_1 e^{-t} \cos 2\sqrt{2}t - 2\sqrt{2}C_1 e^{-t} \sin 2\sqrt{2}t - C_2 e^{-t} \sin 2\sqrt{2}t + 2\sqrt{2}C_2 e^{-t} \cos 2\sqrt{2}t$$

$$y'' = C_1 e^{-t} \cos 2\sqrt{2}t + 2\sqrt{2}C_1 e^{-t} \sin 2\sqrt{2}t + 2\sqrt{2}C_2 e^{-t} \sin 2\sqrt{2}t - 8C_1 e^{-t} \cos 2\sqrt{2}t + C_2 e^{-t} \sin 2\sqrt{2}t - 2\sqrt{2}C_2 e^{-t} \cos 2\sqrt{2}t - 8C_2 e^{-t} \sin 2\sqrt{2}t$$

$$(-7C_1 e^{-t} \cos 2\sqrt{2}t + 4\sqrt{2}C_1 e^{-t} \sin 2\sqrt{2}t - 7C_2 e^{-t} \sin 2\sqrt{2}t - 4\sqrt{2}C_2 e^{-t} \cos 2\sqrt{2}t) +$$

$$(-2C_1 e^{-t} \cos 2\sqrt{2}t - 4\sqrt{2}C_1 e^{-t} \sin 2\sqrt{2}t - 2C_2 e^{-t} \sin 2\sqrt{2}t + 4\sqrt{2}C_2 e^{-t} \cos 2\sqrt{2}t) +$$

$$(9C_1 e^{-t} \cos 2\sqrt{2}t + 9C_2 e^{-t} \sin 2\sqrt{2}t) = 0?$$

$$0 = 0 \quad \checkmark$$

Answer:

$$y = C_1 e^{-t} \cos 2\sqrt{2}t + C_2 e^{-t} \sin 2\sqrt{2}t$$

Points: 5 3 0

4. Solve $y'' - 6y' + 25y = 50t^3 - 36t^2 - 63t + 18$ using the method of undetermined coefficients. You must show all of your work to get full credit.

$$r^2 - 6r + 25 = 0$$

$$r = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(25)}}{2(1)} = \frac{6 \pm 8i}{2} = 3 \pm 4i$$

$$y_h = C_1 e^{3t} \cos 4t + C_2 e^{3t} \sin 4t$$

$$y_p = At^3 + Bt^2 + Ct + D$$

$$y_p' = 3At^2 + 2Bt + C$$

$$y_p'' = 6At + 2B$$

$$(6At + 2B) - 6(3At^2 + 2Bt + C) + 25(At^3 + Bt^2 + Ct + D) = 50t^3 - 36t^2 - 63t + 18$$

$$6At + 2B - 18At^2 - 12Bt - 6C + 25At^3 + 25Bt^2 + 25Ct + 25D =$$

$$25A = 50$$

$$A = 2$$

$$-18A + 25B = -36$$

$$-36 + 25B = -36 \quad B = 0$$

$$6A - 12B + 25C = -63$$

$$12 - 0 + 25C = -63 \quad C = -3$$

$$2B - 6C + 25D = 18$$

$$0 - 6(-3) + 25D = 18$$

$$18 + 25D = 18 \quad D = 0$$

$$y_p = 2t^3 - 3t$$

$$y = y_h + y_p = C_1 e^{3t} \cos 4t + C_2 e^{3t} \sin 4t + 2t^3 - 3t$$

* Check on back *

Answer:

$$y = C_1 e^{3t} \cos 4t + C_2 e^{3t} \sin 4t + 2t^3 - 3t$$

Points: (5) 3 0

5. Solve $y'' - y' - 2y = e^{3t}$ using the method of variation of parameters. You must show all of your work in order to get full credit.

$$r^2 - r - 2 = 0$$

$$(r-2)(r+1) = 0$$

$$r = 2; r = -1$$

$$y_h = C_1 e^{2t} + C_2 e^{-t}$$

$$y_1 = e^{2t} \quad y_1' = 2e^{2t} \quad f = e^{3t}$$

$$y_2 = e^{-t} \quad y_2' = -e^{-t}$$

$$y_p = u_1 y_1 + u_2 y_2$$

$$u_1 = \int \frac{y_2 f}{y_2 y_1' - y_1 y_2'} dt = \int \frac{(e^{-t})(e^{3t})}{(e^{-t})(2e^{2t}) + (e^{2t})(-e^{-t})} dt = \int \frac{e^{3t}}{3e^{2t}} dt = \frac{1}{3} \int e^t dt = \frac{1}{3} e^t$$

$$u_2 = \int \frac{y_1 f}{y_1 y_2' - y_2 y_1'} dt = \int \frac{(e^{2t})(e^{3t})}{(e^{2t})(-e^{-t}) - (e^{-t})(2e^{2t})} dt = \int \frac{e^{3t}}{-3e^{-t}} dt = -\frac{1}{3} \int e^{4t} dt$$

$$= -\frac{1}{3} \int e^u \frac{du}{4} = -\frac{1}{12} e^u = -\frac{1}{12} e^{4t}$$

$u = 4t$
 $\frac{du}{dt} = 4$
 $dt = \frac{du}{4}$

$$y_p = \left(\frac{1}{3} e^t\right)(e^{2t}) + \left(-\frac{1}{12} e^{4t}\right)(e^{-t})$$

$$= \frac{1}{3} e^{3t} - \frac{1}{12} e^{3t} = \frac{1}{4} e^{3t}$$

$$y = y_h + y_p = C_1 e^{2t} + C_2 e^{-t} + \frac{1}{4} e^{3t}$$

Check:

$$y = C_1 e^{2t} + C_2 e^{-t} + \frac{1}{4} e^{3t}$$

$$y' = 2C_1 e^{2t} - C_2 e^{-t} + \frac{3}{4} e^{3t}$$

$$y'' = 4C_1 e^{2t} + C_2 e^{-t} + \frac{9}{4} e^{3t}$$

$$y'' - y' - 2y = e^{3t} ?$$

$$4C_1 e^{2t} + C_2 e^{-t} + \frac{9}{4} e^{3t} - (2C_1 e^{2t} - C_2 e^{-t} + \frac{3}{4} e^{3t})$$

$$- 2(C_1 e^{2t} + C_2 e^{-t} + \frac{1}{4} e^{3t}) = e^{3t} ?$$

$$4C_1 e^{2t} + C_2 e^{-t} + \frac{9}{4} e^{3t} - 2C_1 e^{2t} - C_2 e^{-t} - \frac{3}{4} e^{3t}$$

$$- 2C_1 e^{2t} - C_2 e^{-t} - \frac{1}{2} e^{3t} = e^{3t} ?$$

$$\frac{9}{4} e^{3t} - \frac{3}{4} e^{3t} - \frac{1}{2} e^{3t} = e^{3t} \quad \checkmark$$

Answer:

$$y = C_1 e^{2t} + C_2 e^{-t} + \frac{1}{4} e^{3t}$$

Points: (5) 3 0

$$my'' + \gamma y' + ky = f(t)$$

$\frac{1}{2}$ foot

$\frac{1}{3}$ foot

6. A 5-pound body attached to an elastic helical spring stretches it 4 inches. After it comes to rest, it is stretched an additional 6 inches and released. Find its equation of motion using seconds for the unit of time. Use $g = 32 \text{ ft/s}^2$ for the gravity constant.

$$W = 5 \text{ lb} \quad l = 4'' = \frac{1}{3}'$$

$$m = \frac{5}{32} \quad K = \frac{W}{l} = \frac{5}{\frac{1}{3}} = 15$$

$$y(0) = 6'' = \frac{1}{2}'$$

$$y'(0) = 0$$

$$f(t) = 0$$

$$\frac{5}{32}y'' + 15y = 0$$

$$y'' + 96y = 0$$

$$r^2 + 96 = 0$$

$$r = \pm 4\sqrt{6}i$$

$$y = C_1 e^{0t} \cos 4\sqrt{6}t + C_2 e^{0t} \sin 4\sqrt{6}t$$

$$y = C_1 \cos 4\sqrt{6}t + C_2 \sin 4\sqrt{6}t$$

$$y(0) = \frac{1}{2} = C_1$$

$$y' = -4\sqrt{6}C_1 \sin 4\sqrt{6}t + 4\sqrt{6}C_2 \cos 4\sqrt{6}t$$

$$y'(0) = 0 = 4\sqrt{6}C_2$$

$$C_2 = 0$$

$$y = \frac{1}{2} \cos 4\sqrt{6}t$$

Check:

$$y = C_1 \cos 4\sqrt{6}t + C_2 \sin 4\sqrt{6}t$$

$$y' = -4\sqrt{6}C_1 \sin 4\sqrt{6}t + 4\sqrt{6}C_2 \cos 4\sqrt{6}t$$

$$y'' = -96C_1 \cos 4\sqrt{6}t - 96C_2 \sin 4\sqrt{6}t$$

$$\frac{5}{32}(-96C_1 \cos 4\sqrt{6}t - 96C_2 \sin 4\sqrt{6}t) + 15C_1 \cos 4\sqrt{6}t + 15C_2 \sin 4\sqrt{6}t = 0?$$

$$-15C_1 \cos 4\sqrt{6}t - 15C_2 \sin 4\sqrt{6}t + 15C_1 \cos 4\sqrt{6}t + 15C_2 \sin 4\sqrt{6}t = 0?$$

$$0 = 0 \checkmark$$

$$y = \frac{1}{2} \cos 4\sqrt{6}t$$

$$y' = -2\sqrt{6} \sin 4\sqrt{6}t$$

$$y'' = -48 \cos 4\sqrt{6}t$$

$$\frac{5}{32}(-48 \cos 4\sqrt{6}t) + 15(\frac{1}{2} \cos 4\sqrt{6}t) = 0?$$

$$-\frac{15}{2} \cos 4\sqrt{6}t + \frac{15}{2} \cos 4\sqrt{6}t = 0?$$

$$0 = 0 \checkmark$$

Answer:

$$y = \frac{1}{2} \cos 4\sqrt{6}t$$

Points: (5) 3 0

$$my'' + \gamma y' + ky = f(t)$$

7. A 16-lb weight stretches a spring $\frac{1}{2}$ foot. A forcing function $f(t) = 10 \sin 2t$ is attached to the system. Find the equation of motion. Use $g = 32 \text{ ft/s}^2$ for the gravity constant.

$$W = 16 \text{ lb} \quad f(t) = 10 \sin 2t$$

$$m = \frac{16}{32} = \frac{1}{2} \quad l = \frac{1}{2}; K = \frac{16}{\frac{1}{2}} = 32$$

$$y(0) = 0 \quad y'(0) = 0$$

$$\frac{1}{2}y'' + 32y = 10 \sin 2t$$

$$y'' + 64y = 20 \sin 2t$$

$$y_p = A \sin 2t + B \cos 2t$$

$$y'_p = 2A \cos 2t - 2B \sin 2t$$

$$y''_p = -4A \sin 2t - 4B \cos 2t$$

$$-4A \sin 2t - 4B \cos 2t + 64A \sin 2t + 64B \cos 2t = 20 \sin 2t$$

$$60A = 20$$

$$A = \frac{1}{3}$$

$$60B = 0 \quad B = 0$$

$$y_p = \frac{1}{3} \sin 2t$$

$$y = y_h + y_p = C_1 \cos 8t + C_2 \sin 8t + \frac{1}{3} \sin 2t$$

$$y' = -8C_1 \sin 8t + 8C_2 \cos 8t + \frac{2}{3} \cos 2t$$

$$y(0) = 0 = C_1$$

$$y'(0) = 0 = 8C_2 + \frac{2}{3}$$

$$C_2 = -\frac{1}{12}$$

$$y = -\frac{1}{12} \sin 8t + \frac{1}{3} \sin 2t$$

Check:

$$y'' = -64C_1 \cos 8t - 64C_2 \sin 8t - \frac{1}{3} \sin 2t$$

$$\frac{1}{2}(-64C_1 \cos 8t - 64C_2 \sin 8t - \frac{1}{3} \sin 2t)$$

$$+ 32(C_1 \cos 8t + C_2 \sin 8t + \frac{1}{3} \sin 2t) = 10 \sin 2t$$

$$-\frac{2}{3} + \frac{32}{3} = \frac{30}{3} = 10 \checkmark$$

Check:

$$y = -\frac{1}{12} \sin 8t + \frac{1}{3} \sin 2t$$

$$y' = -\frac{8}{3} \cos 8t + \frac{2}{3} \cos 2t$$

$$y'' = \frac{16}{3} \sin 8t - \frac{4}{3} \sin 2t$$

$$\frac{8}{3} \sin 8t - \frac{2}{3} \sin 2t + \frac{8}{3} \sin 8t + \frac{32}{3} \sin 2t = 10 \sin 2t ?$$

$$-\frac{2}{3} + \frac{32}{3} = 10 \checkmark$$

Answer:

$$y = -\frac{1}{12} \sin 8t + \frac{1}{3} \sin 2t$$

Points: (5) 3 0

$$my'' + \gamma y' + ky = f(t)$$

→ 8/3 feet

8. A spring is stretched 32 inches by an object weighing 2 pounds, and brought to rest. It is then given an additional pull of 1 foot and released. If the spring is immersed in a medium whose coefficient of resistance is $\gamma = \frac{1}{2}$, find the equation of motion of the object. Use $g = 32 \text{ ft/s}^2$ for the gravity constant.

$$W = 2 \quad l = \frac{8}{3}$$

$$m = \frac{2}{32} = \frac{1}{16} \quad k = \frac{2}{8/3} = \frac{3}{4}$$

$$y(0) = 1 \quad \gamma = \frac{1}{2}$$

$$y'(0) = 0 \quad f(t) = 0$$

$$\frac{1}{16}y'' + \frac{1}{2}y' + \frac{3}{4}y = 0$$

$$y'' + 8y' + 12y = 0$$

$$(r+6)(r+2) = 0$$

$$r = -6; r = -2$$

$$y = C_1 e^{-6t} + C_2 e^{-2t}$$

Check:

$$y = C_1 e^{-6t} + C_2 e^{-2t}$$

$$y' = -6C_1 e^{-6t} - 2C_2 e^{-2t}$$

$$y'' = 36C_1 e^{-6t} + 4C_2 e^{-2t}$$

$$\frac{1}{16}(36C_1 e^{-6t} + 4C_2 e^{-2t}) + \frac{1}{2}(-6C_1 e^{-6t} - 2C_2 e^{-2t}) + \frac{3}{4}(C_1 e^{-6t} + C_2 e^{-2t}) = 0?$$

$$\frac{9}{4} - 3 + \frac{3}{4} = 0 \checkmark$$

$$\frac{1}{4} - 1 + \frac{3}{4} = 0 \checkmark$$

$$y(0) = 1 = C_1 + C_2$$

$$y'(0) = 0 = -6C_1 - 2C_2$$

$$C_1 = 1 - C_2$$

$$0 = -6(1 - C_2) - 2C_2$$

$$= -6 + 6C_2 - 2C_2$$

$$= -6 + 4C_2$$

$$C_2 = \frac{3}{2}$$

$$C_1 = 1 - \frac{3}{2} = -\frac{1}{2}$$

$$y = -\frac{1}{2}e^{-6t} + \frac{3}{2}e^{-2t}$$

Check:

$$y = -\frac{1}{2}e^{-6t} + \frac{3}{2}e^{-2t}$$

$$y' = 3e^{-6t} - 3e^{-2t}$$

$$y'' = -18e^{-6t} + 6e^{-2t}$$

$$\frac{1}{16}(-18e^{-6t} + 6e^{-2t}) + \frac{1}{2}(3e^{-6t} - 3e^{-2t}) + \frac{3}{4}(-\frac{1}{2}e^{-6t} + \frac{3}{2}e^{-2t}) = 0?$$

$$-\frac{9}{8} + \frac{3}{2} - \frac{3}{8} = 0 \checkmark$$

$$\frac{3}{8} - \frac{3}{2} + \frac{9}{8} = 0 \checkmark$$

Answer:

$$y = -\frac{1}{2}e^{-6t} + \frac{3}{2}e^{-2t}$$

Points: 5 3 0

$$my'' + \gamma y' + ky = 0$$

9. A forcing function $f(t) = \frac{5}{2} \cos 2t$ is applied to the motion in question #8. Find the equation of motion. Use $g = 32 \text{ ft/s}^2$ for the gravity constant.

from question 8: $\frac{1}{6}y'' + \frac{1}{2}y' + \frac{3}{4}y = 0$; $y(0) = 1$; $y'(0) = 0$

$$y_h = C_1 e^{-4t} + C_2 e^{-2t}$$

$$\frac{1}{6}y'' + \frac{1}{2}y' + \frac{3}{4}y = \frac{5}{2} \cos 2t$$

$$y_p = A \cos 2t + B \sin 2t$$

$$y'_p = -2A \sin 2t + 2B \cos 2t$$

$$y''_p = -4A \cos 2t - 4B \sin 2t$$

$$\frac{1}{6}(-4A \cos 2t - 4B \sin 2t) + \frac{1}{2}(-2A \sin 2t + 2B \cos 2t) + \frac{3}{4}(A \cos 2t + B \sin 2t) = \frac{5}{2} \cos 2t$$

$$-\frac{1}{4}A \cos 2t - \frac{1}{4}B \sin 2t - A \sin 2t + B \cos 2t + \frac{3}{4}A \cos 2t + \frac{3}{4}B \sin 2t = \frac{5}{2} \cos 2t$$

$$-\frac{1}{4}A + B + \frac{3}{4}A = \frac{5}{2}$$

$$-\frac{1}{4}B - A + \frac{3}{4}B = 0$$

$$-A + \frac{1}{2}B = 0 \quad A = \frac{1}{2}B$$

$$\frac{1}{2}A + B = \frac{5}{2}$$

$$\frac{1}{2}(\frac{1}{2}B) + B = \frac{5}{2}$$

$$\frac{1}{4}B + B = \frac{5}{2}$$

$$\frac{5}{4}B = \frac{5}{2}$$

$$B = 2$$

$$A = 1$$

$$y_p = \cos 2t + 2 \sin 2t$$

$$y = C_1 e^{-4t} + C_2 e^{-2t} + \cos 2t + 2 \sin 2t$$

$$y' = -4C_1 e^{-4t} - 2C_2 e^{-2t} - 2 \sin 2t + 4 \cos 2t$$

* Check on back *

$$y(0) = C_1 + C_2 + 1 = 1$$

$$y'(0) = -4C_1 - 2C_2 + 4 = 0$$

$$C_1 = -C_2$$

$$-4(-C_2) - 2C_2 + 4 = 0$$

$$6C_2 - 2C_2 = -4$$

$$4C_2 = -4$$

$$C_2 = -1$$

$$C_1 = 1$$

$$y = e^{-4t} - e^{-2t} + \cos 2t + 2 \sin 2t$$

Answer:

$$y = e^{-4t} - e^{-2t} + \cos 2t + 2 \sin 2t$$

Points: 5 3 0

Check:

$$y = C_1 e^{-6t} + C_2 e^{-2t} + \cos 2t + 2 \sin 2t$$

$$y' = -6C_1 e^{-6t} - 2C_2 e^{-2t} - 2 \sin 2t + 4 \cos 2t$$

$$y'' = 36C_1 e^{-6t} + 4C_2 e^{-2t} - 4 \cos 2t - 8 \sin 2t$$

$$\frac{1}{6} y'' + \frac{1}{2} y' + \frac{3}{4} y = \frac{5}{2} \cos 2t$$

* Already proved homogeneous from question 8*

$$\frac{1}{6} (-4 \cos 2t - 8 \sin 2t) + \frac{1}{2} (-2 \sin 2t + 4 \cos 2t) + \frac{3}{4} (\cos 2t + 2 \sin 2t) = \frac{5}{2} \cos 2t ?$$

$$-\frac{1}{4} + 2 + \frac{3}{4} = \frac{5}{2} \checkmark$$

$$-\frac{1}{2} - 1 + \frac{3}{2} = 0 \checkmark$$

$$y = e^{-6t} - e^{-2t} + \cos 2t + 2 \sin 2t$$

$$y' = -6e^{-6t} + 2e^{-2t} + \dots$$

$$y'' = 36e^{-6t} - 4e^{-2t} + \dots$$

* Already proved particular *

$$\frac{1}{6} (36e^{-6t} - 4e^{-2t}) + \frac{1}{2} (-6e^{-6t} + 2e^{-2t}) + \frac{3}{4} (e^{-6t} - e^{-2t}) = 0 ?$$

$$\frac{9}{4} - 3 + \frac{3}{4} = 0 \checkmark$$

$$-\frac{1}{4} + 1 - \frac{3}{4} = 0 \checkmark$$

yay!

Math 316 - Exam #4

Name: Seth Ricks (2 points will be lost for not writing your name)

Score: 54 / 60 = 90 %

1. Find the eigenvalues and associated eigenvectors of the matrix

$$A = \begin{bmatrix} 4 & 2 \\ 3 & -1 \end{bmatrix}$$

Write the answer in the space below in the form: λ = eigenvalue, eigenvector. You must show your work in order to get full credit.

$$A = \begin{bmatrix} 4 & 2 \\ 3 & -1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 4-\lambda & 2 \\ 3 & -1-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = (4-\lambda)(-1-\lambda) - (3)(2) = 0$$

$$= -4 - 4\lambda + \lambda + \lambda^2 - 6 = 0$$

$$= \lambda^2 - 3\lambda - 10 = 0$$

$$(\lambda - 5)(\lambda + 2) = 0$$

$$\lambda = 5; -2$$

Check:

$$A \cdot V_1 = \lambda_1 \cdot V_1 ?$$

$$\begin{bmatrix} 4 & 2 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix} = 5 \begin{bmatrix} 2 \\ 1 \end{bmatrix} \checkmark$$

$$A \cdot V_2 = \lambda_2 \cdot V_2 ?$$

$$\begin{bmatrix} 4 & 2 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ -3 \end{bmatrix} = \begin{bmatrix} -2 \\ 6 \end{bmatrix} = -2 \begin{bmatrix} 1 \\ -3 \end{bmatrix} \checkmark$$

$$\lambda_1 = 5, \begin{bmatrix} 2 \\ 1 \end{bmatrix}; \lambda_2 = -2, \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

Answer:

$$\lambda_1 = 5, \begin{bmatrix} 2 \\ 1 \end{bmatrix} \checkmark$$

$$\lambda_2 = -2, \begin{bmatrix} 1 \\ -3 \end{bmatrix} \checkmark$$

Points: 5 3 0

2. Solve the system of differential equations using the eigenvalue/eigenvector method. You must show all of your work in order to receive full credit.

$$\begin{aligned}x_1' &= 2x_1 + 3x_2 \\x_2' &= 2x_1 + x_2\end{aligned}$$

$$X' = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} X$$

$$A - \lambda I = \begin{bmatrix} 2-\lambda & 3 \\ 2 & 1-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = (2-\lambda)(1-\lambda) - 3(2) = 0$$

$$= 2 - 2\lambda - \lambda + \lambda^2 - 6 = 0$$

$$= \lambda^2 - 3\lambda - 4 = 0$$

$$(\lambda - 4)(\lambda + 1) = 0$$

$$\lambda = 4; -1$$

$$\lambda = 4$$

$$\begin{bmatrix} 2-4 & 3 & 0 \\ 2 & 1-4 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} -2 & 3 & 0 \\ 2 & -3 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -2 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{aligned} -2V_1 + 3V_2 &= 0 \\ V_1 &= \frac{3}{2}V_2 \end{aligned}$$

$$V_1 = \begin{bmatrix} \frac{3}{2} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{2}{3} \end{bmatrix} \quad V_2 = V_2$$

$$\lambda = -1$$

$$\begin{bmatrix} 2+1 & 3 & 0 \\ 2 & 1+1 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 3 & 3 & 0 \\ 2 & 2 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{aligned} 3V_1 &= -3V_2 \\ V_1 &= -V_2 \\ V_2 &= V_2 \end{aligned}$$

$$V_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Check:

$$A \cdot V_1 = \lambda_1 \cdot V_1?$$

$$\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ \frac{2}{3} \end{bmatrix} = \begin{bmatrix} 4 \\ \frac{8}{3} \end{bmatrix} = 4 \begin{bmatrix} 1 \\ \frac{2}{3} \end{bmatrix} \checkmark$$

$$A \cdot V_2 = \lambda_2 \cdot V_2?$$

$$\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} = -1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} \checkmark$$

$$X = C_1 e^{\lambda_1 t} V_1 + C_2 e^{\lambda_2 t} V_2$$

$$X = C_1 e^{4t} \begin{bmatrix} 1 \\ \frac{2}{3} \end{bmatrix} + C_2 e^{-t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

* Check on back.

Answer:

$$X = C_1 e^{4t} \begin{bmatrix} 1 \\ \frac{2}{3} \end{bmatrix} + C_2 e^{-t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Points: 5 3 0

$$x = C_1 e^{4t} \begin{bmatrix} 1 \\ 2/3 \end{bmatrix} + C_2 e^{-t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$x' = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} x$$

$$x = \begin{bmatrix} C_1 e^{4t} + C_2 e^{-t} \\ \frac{2}{3} C_1 e^{4t} - C_2 e^{-t} \end{bmatrix}$$

$$x' = \begin{bmatrix} 4C_1 e^{4t} - C_2 e^{-t} \\ \frac{8}{3} C_1 e^{4t} + C_2 e^{-t} \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} C_1 e^{4t} + C_2 e^{-t} \\ \frac{2}{3} C_1 e^{4t} - C_2 e^{-t} \end{bmatrix}$$

$$= \begin{bmatrix} 2(C_1 e^{4t} + C_2 e^{-t}) + 3(\frac{2}{3} C_1 e^{4t} - C_2 e^{-t}) \\ 2(\underbrace{\phantom{C_1 e^{4t} + C_2 e^{-t}}} \downarrow) + 1(\underbrace{\phantom{\frac{2}{3} C_1 e^{4t} - C_2 e^{-t}}} \downarrow) \end{bmatrix}$$

$$= \begin{bmatrix} 4C_1 e^{4t} - C_2 e^{-t} \\ \frac{8}{3} C_1 e^{4t} + C_2 e^{-t} \end{bmatrix} = x' \quad \checkmark \checkmark$$

3. Solve the system of differential equations using the eigenvalue/eigenvector method. You must show all of your work in order to receive full credit.

$$x' = \begin{bmatrix} 1 & -3 \\ 3 & 7 \end{bmatrix} x$$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & -3 \\ 3 & 7-\lambda \end{bmatrix}$$

$$\begin{aligned} \det(A - \lambda I) &= (1-\lambda)(7-\lambda) + 9 = 0 \\ &= 7 - \lambda - 7\lambda + \lambda^2 + 9 = 0 \\ &= \lambda^2 - 8\lambda + 16 = 0 \\ &= (\lambda - 4)^2 = 0 \\ \lambda &= 4; 4 \end{aligned}$$

$$\lambda = 4$$

$$\begin{bmatrix} -3 & -3 & 0 \\ 3 & 3 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} -v_1 - v_2 &= 0 \\ v_1 &= -v_2 \\ v_2 &= v_2 \end{aligned}$$

$$v_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$A v_2 = v_1$$

$$\begin{bmatrix} -3 & -3 & 1 \\ 3 & 3 & -1 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} -v_1 - v_2 &= 1 \\ -v_1 &= v_2 + 1 \\ v_1 &= -v_2 - 1 \\ v_2 &= v_2 \\ v_2 &= \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} x_1' &= x_1 - 3x_2 \\ x_2' &= 3x_1 + 7x_2 \end{aligned}$$

$$x = C_1 e^{4t} \begin{bmatrix} 1 \\ -1 \end{bmatrix} + C_2 e^{4t} \left(\begin{bmatrix} 1 \\ -1 \end{bmatrix} t + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right)$$

Check:

$$x = \begin{bmatrix} C_1 e^{4t} + C_2 e^{4t} t + C_2 e^{4t} \\ -C_1 e^{4t} - C_2 e^{4t} t - C_2 e^{4t} \end{bmatrix}$$

$$x' = \begin{bmatrix} 4C_1 e^{4t} + 4C_2 e^{4t} t + C_2 e^{4t} + 4C_2 e^{4t} \\ -4C_1 e^{4t} - 4C_2 e^{4t} t - C_2 e^{4t} - 4C_2 e^{4t} \end{bmatrix}$$

$$\begin{aligned} A \cdot x &= \begin{bmatrix} 1(C_1 e^{4t} + C_2 e^{4t} t + C_2 e^{4t}) - 3(-C_1 e^{4t} - C_2 e^{4t} t - C_2 e^{4t}) \\ 3(C_1 e^{4t} + C_2 e^{4t} t + C_2 e^{4t}) + 7(-C_1 e^{4t} - C_2 e^{4t} t - C_2 e^{4t}) \end{bmatrix} \\ &= \begin{bmatrix} 4C_1 e^{4t} + 4C_2 e^{4t} t + C_2 e^{4t} + 4C_2 e^{4t} \\ -4C_1 e^{4t} - 4C_2 e^{4t} t - C_2 e^{4t} - 4C_2 e^{4t} \end{bmatrix} \end{aligned}$$

Answer:

$$x = C_1 e^{4t} \begin{bmatrix} 1 \\ -1 \end{bmatrix} + C_2 e^{4t} \left(\begin{bmatrix} 1 \\ -1 \end{bmatrix} t + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right)$$

Points: 5 3 0

4. Solve the system of differential equations using the eigenvalue/vector method. You must show all of your work in order to receive full credit.

$$X' = \begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix} X$$

$$A - \lambda I = (4 - \lambda)(4 - \lambda) + 9 = 0$$

$$= 16 - 8\lambda + \lambda^2 + 9 = 0$$

$$= \lambda^2 - 8\lambda + 25 = 0$$

$$\lambda = \frac{-(-8) \pm \sqrt{(-8)^2 - 4(1)(25)}}{2(1)}$$

$$= \frac{8 \pm \sqrt{64 - 100}}{2} = 4 \pm 3i$$

$$\begin{bmatrix} 4 - (4 + 3i) & -3 & 0 \\ 3 & 4 - (4 + 3i) & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -3i & -3 & 0 \\ 3 & -3i & 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -3i & 0 \\ 3 & -3i & 0 \end{bmatrix}$$

$$3v_1 - 3iv_2 = 0$$

$$3v_1 = 3iv_2$$

$$v_1 = iv_2$$

$$v_2 = v_2$$

$$v_1 = \begin{bmatrix} i \\ 1 \end{bmatrix}$$

$$x'_1 = 4x_1 - 3x_2$$

$$x'_2 = 3x_1 + 4x_2$$

$$e^{4t} \begin{bmatrix} i \\ 1 \end{bmatrix} \begin{bmatrix} \cos 3t + i \sin 3t \end{bmatrix}$$

$$= e^{4t} \begin{bmatrix} i \cos 3t - \sin 3t \\ \cos 3t + i \sin 3t \end{bmatrix}$$

$$= e^{4t} \begin{bmatrix} -\sin 3t & \cos 3t \\ \cos 3t & \sin 3t \end{bmatrix}$$

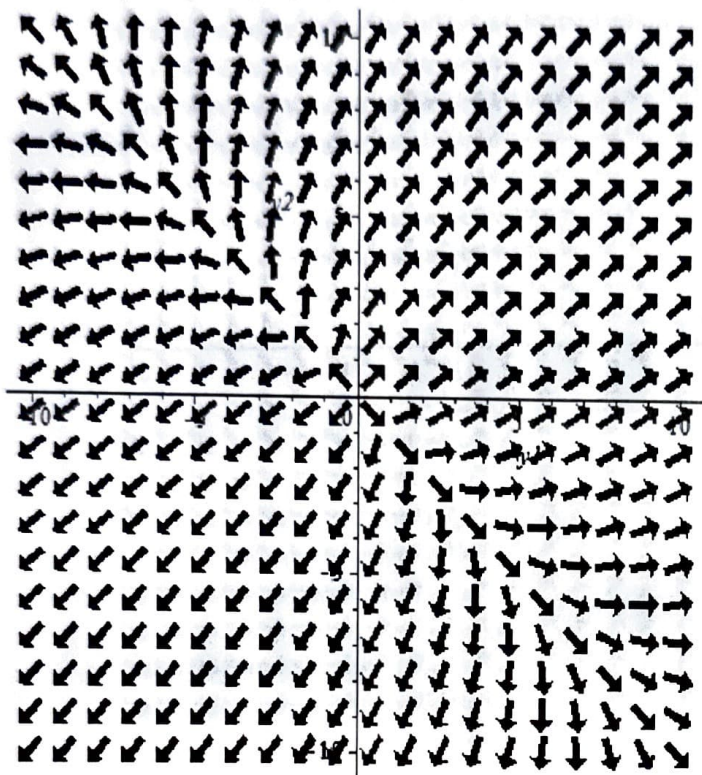
$$X = C_1 e^{4t} \begin{bmatrix} -\sin 3t \\ \cos 3t \end{bmatrix} + C_2 e^{4t} \begin{bmatrix} \cos 3t \\ \sin 3t \end{bmatrix}$$

Answer:

$$X = C_1 e^{4t} \begin{bmatrix} -\sin 3t \\ \cos 3t \end{bmatrix} + C_2 e^{4t} \begin{bmatrix} \cos 3t \\ \sin 3t \end{bmatrix}$$

Points: 5 3 0

5. Using the given phase portrait, determine the stability at the critical point $x = 0$ (stable or unstable) and the type of node at the critical point.



- Repeated positive eigenvalues
- Unstable
- Proper node (star point)

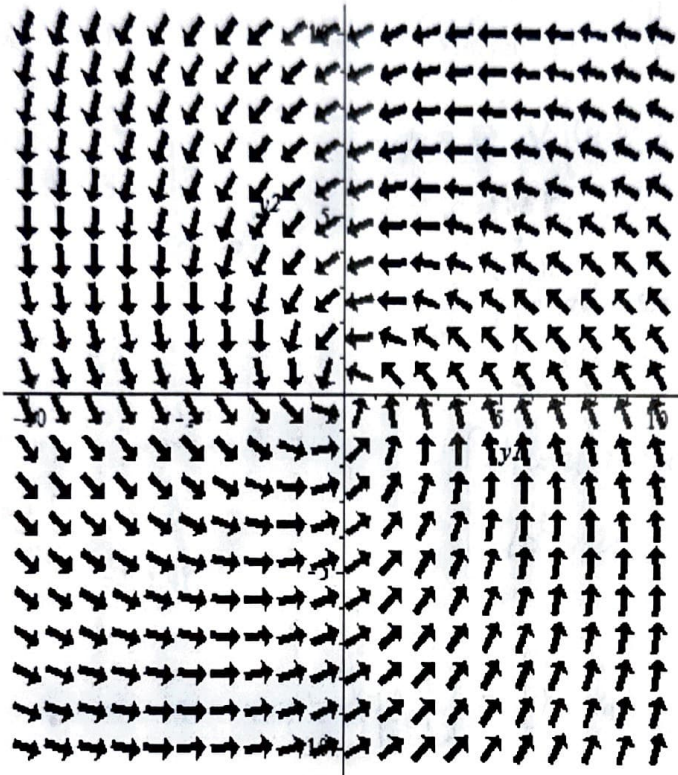
Answer:

Stability - unstable

Type of Node - ~~Proper Node~~
(star point)

Points: 5 3 0

6. Using the given phase portrait, determine the stability at the critical point $x = 0$ (stable or unstable) and the type of node at the critical point.



- Complex eigenvalues
- $\alpha < 0$
- ⑥ • Stable
- Spiral point

Answer:

Stability - Stable

Type of Node - Spiral point

Points: 5 3 0

7. Find the solution satisfying the initial conditions for

$$x' = \begin{bmatrix} 7 & -5 \\ 4 & 3 \end{bmatrix} x, x(0) = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

using the Exponential matrix method. You must show all of your work in order to receive full credit.

$$A - \lambda I = \begin{bmatrix} 7 - \lambda & -5 \\ 4 & 3 - \lambda \end{bmatrix}$$

$$\det(A - \lambda I) = (7 - \lambda)(3 - \lambda) + 20 = 0$$

$$= 21 - 7\lambda - 3\lambda + \lambda^2 + 20 = 0$$

$$= \lambda^2 - 10\lambda + 41 = 0$$

$$= \frac{-(-10) \pm \sqrt{(-10)^2 - 4(1)(41)}}{2(1)}$$

$$= \frac{10 \pm \sqrt{64}}{2(1)} = 5 \pm \frac{8i}{2}$$

$$= 5 \pm 4i$$

$$\lambda = 5 + 4i$$

$$\begin{bmatrix} 7 - (5 + 4i) & -5 & 0 \\ 4 & 3 - (5 + 4i) & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2 - 4i & -5 & 0 \\ 4 & -2 - 4i & 0 \end{bmatrix}$$

$$(2 + 4i)(2 - 4i) = (2)(2) + 2(4i) + 2(4i) + (4i)(-4i)$$

$$= 4 + 16 = 20$$

$$\begin{bmatrix} 20 & -5(2 + 4i) & 0 \\ 4 & -2 - 4i & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 20 & -10 - 20i & 0 \\ 4 & -2 - 4i & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -1 - 2i & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$2V_1 + (-1 - 2i)V_2 = 0$$

$$V_1 = \frac{1}{2} + iV_2$$

$$V_2 = V_2$$

$$V = \begin{bmatrix} \frac{1}{2} + i \\ 1 \end{bmatrix} = \begin{bmatrix} 1 + 2i \\ 2 \end{bmatrix} \quad P = \begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix}$$

$$\phi = e^{at} v \begin{bmatrix} \cos bt + i \sin bt \end{bmatrix}$$

$$= e^t \begin{bmatrix} 1 + 2i \\ 2 \end{bmatrix} \begin{bmatrix} \cos 2t + i \sin 2t \end{bmatrix}$$

$$= e^t \begin{bmatrix} \cos 2t - 2 \sin 2t + 2i \cos 2t + i \sin 2t \\ 2 \cos 2t + 2i \sin 2t \end{bmatrix}$$

$$= \begin{bmatrix} e^t (\cos 2t - 2 \sin 2t) & e^t (2 \cos 2t + i \sin 2t) \\ 2e^t \cos 2t & 2e^t i \sin 2t \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix}$$

$$P^{-1} = \frac{1}{-4} \begin{bmatrix} 0 & -2 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{4} \end{bmatrix}$$

$$\begin{bmatrix} e^t \cos 2t - 2e^t \sin 2t & 2e^t \cos 2t + e^t i \sin 2t \\ 2e^t \cos 2t & 2e^t i \sin 2t \end{bmatrix} \cdot \begin{bmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{4} \end{bmatrix}$$

$$= \begin{bmatrix} 0 + \frac{1}{2}(2e^t \cos 2t + e^t i \sin 2t) & \frac{1}{2}(e^t \cos 2t - 2e^t i \sin 2t) - \frac{1}{4}(2e^t \cos 2t + e^t i \sin 2t) \\ 0 + e^t i \sin 2t & \frac{1}{2}(2e^t \cos 2t) - \frac{1}{4}(2e^t i \sin 2t) \end{bmatrix}$$

$$= \begin{bmatrix} e^t \cos 2t + \frac{1}{2}e^t i \sin 2t & -\frac{1}{4}e^t i \sin 2t \\ e^t i \sin 2t & e^t \cos 2t - \frac{1}{2}e^t i \sin 2t \end{bmatrix}$$

$$\begin{bmatrix} \uparrow \\ \uparrow \end{bmatrix} \cdot \begin{bmatrix} ? \\ 0 \end{bmatrix} = \begin{bmatrix} 2(e^t \cos 2t + \frac{1}{2}e^t i \sin 2t) \\ 2(e^t i \sin 2t) \end{bmatrix} = \begin{bmatrix} 2e^t \cos 2t + e^t i \sin 2t \\ 2e^t i \sin 2t \end{bmatrix}$$

Answer:

$$X = \begin{bmatrix} 2e^t \cos 2t + e^t i \sin 2t \\ 2e^t i \sin 2t \end{bmatrix}$$

Points: 5 (3) 0

8. Transform the second-order differential equation

$$x'' + 3x' + 7x = t^2$$

into a system of two first order differential.

$$\begin{aligned}y &= x_1 \\ y' &= x_2 \\ x_1' &= x_2\end{aligned}$$

$$x_2' + 3x_2 + 7x_1 = t^2$$

$$x_2' = -7x_1 - 3x_2 + t^2$$

$$x' = \begin{bmatrix} 0 & 1 \\ -7 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ t^2 \end{bmatrix}$$

Answer:

$$x' = \begin{bmatrix} 0 & 1 \\ -7 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ t^2 \end{bmatrix}$$

Points: 5 3 0

9. Find only the particular solution, x_p , for the system of nonhomogeneous linear system of differential equations using the method of undetermined coefficients

$$x_1' = 3x_1 + 2x_2 + 3$$

$$x_2' = 7x_1 + 5x_2 + 2t$$

$$X' = \begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix} X + \begin{bmatrix} 3 \\ 2t \end{bmatrix}$$

$$X_p = \begin{bmatrix} At+B \\ Ct+D \end{bmatrix}$$

$$X_p' = \begin{bmatrix} A \\ C \end{bmatrix}$$

$$\begin{bmatrix} A \\ C \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix} \begin{bmatrix} At+B \\ Ct+D \end{bmatrix} + \begin{bmatrix} 3 \\ 2t \end{bmatrix}$$

$$= \begin{bmatrix} 3(At+B) + 2(Ct+D) + 3 \\ 7(At+B) + 5(Ct+D) + 2t \end{bmatrix}$$

$$= \begin{bmatrix} 3At + 2Ct + 3B + 2D + 3 \\ 7At + 5Ct + 7B + 5D + 2t \end{bmatrix}$$

$$A = 3B + 2D + 3$$

$$0 = 3A + 2C$$

$$C = 7B + 5D$$

$$0 = 7A + 5C + 2$$

$$A - 3B + 0C - 2D = 3$$

$$3A + 0B + 2C + 0D = 0$$

$$0A - 7B + C - 5D = 0$$

$$7A + 0B + 5C + 0D = -2$$

$$\begin{bmatrix} 1 & -3 & 0 & -2 & 3 \\ 3 & 0 & 2 & 0 & 0 \\ 0 & -7 & 1 & -5 & 0 \\ 7 & 0 & 5 & 0 & -2 \end{bmatrix}$$

$$\begin{array}{l} -3R_1 + R_2 \Rightarrow R_2 \\ -7R_1 + R_4 \Rightarrow R_4 \end{array} \begin{bmatrix} 1 & -3 & 0 & -2 & 3 \\ 0 & 9 & 2 & 6 & -9 \\ 0 & -7 & 1 & -5 & 0 \\ 0 & 21 & 5 & 14 & -23 \end{bmatrix}$$

$$\frac{1}{9}R_2 \Rightarrow R_2 \begin{bmatrix} 1 & -3 & 0 & -2 & 3 \\ 0 & 1 & \frac{2}{9} & \frac{2}{3} & -1 \\ 0 & -7 & 1 & -5 & 0 \\ 0 & 21 & 5 & 14 & -23 \end{bmatrix}$$

$$\begin{array}{l} 3R_2 + R_1 \Rightarrow R_1 \\ 7R_2 + R_3 \Rightarrow R_3 \\ -21R_2 + R_4 \Rightarrow R_4 \end{array} \begin{bmatrix} 1 & 0 & \frac{4}{3} & 0 & 0 \\ 0 & 1 & \frac{2}{9} & \frac{2}{3} & -1 \\ 0 & 0 & \frac{23}{9} & -\frac{1}{3} & -7 \\ 0 & 0 & \frac{1}{3} & 0 & -2 \end{bmatrix}$$

$$\frac{9}{23}R_3 \Rightarrow R_3 \begin{bmatrix} 1 & 0 & \frac{4}{3} & 0 & 0 \\ 0 & 1 & \frac{2}{9} & \frac{2}{3} & -1 \\ 0 & 0 & 1 & -\frac{3}{23} & -\frac{63}{23} \\ 0 & 0 & \frac{1}{3} & 0 & -2 \end{bmatrix}$$

$$\begin{array}{l} -\frac{2}{3}R_3 + R_1 \Rightarrow R_1 \\ -\frac{2}{9}R_3 + R_2 \Rightarrow R_2 \\ -\frac{1}{3}R_3 + R_4 \Rightarrow R_4 \end{array} \begin{bmatrix} 1 & 0 & 0 & \frac{2}{23} & \frac{42}{23} \\ 0 & 1 & 0 & \frac{16}{23} & -\frac{9}{23} \\ 0 & 0 & 1 & -\frac{3}{23} & -\frac{63}{23} \\ 0 & 0 & 0 & \frac{1}{23} & -\frac{25}{23} \end{bmatrix}$$

$$23R_4 \Rightarrow R_4$$

$$\begin{bmatrix} 1 & 0 & 0 & \frac{2}{23} & \frac{42}{23} \\ 0 & 1 & 0 & \frac{16}{23} & -\frac{9}{23} \\ 0 & 0 & 1 & -\frac{3}{23} & -\frac{63}{23} \\ 0 & 0 & 0 & 1 & -25 \end{bmatrix}$$

$$\frac{3}{23}R_4 + R_3 \Rightarrow R_3$$

$$-\frac{16}{23}R_4 + R_2 \Rightarrow R_2$$

$$-\frac{2}{23}R_4 + R_1 \Rightarrow R_1$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 & 17 \\ 0 & 0 & 1 & 0 & -6 \\ 0 & 0 & 0 & 1 & -25 \end{bmatrix}$$

$$A = 4$$

$$B = 17$$

$$C = -6$$

$$D = -25$$

Answer:

$$X_p = \begin{bmatrix} 4t + 17 \\ -6t - 25 \end{bmatrix}$$

Points: 5 3 0

* Check on back

10. Find only the particular solution, x_p , the system of nonhomogeneous linear system of differential equations using the method of variation of parameters

$$x_1' = -5x_1 + 1x_2 + 6e^{2t}$$

$$x_2' = 4x_1 - 2x_2 - e^{2t}$$

$$A = \begin{bmatrix} -5 & 1 \\ 4 & -2 \end{bmatrix}$$

Given that the homogeneous solution, x_h , is

$$x_h = c_1 e^{-t} \begin{bmatrix} 1 \\ 4 \end{bmatrix} + c_2 e^{-6t} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$x_p = \Phi \cdot \int \Phi^{-1} \cdot b \, dt$$

$$\Phi = [e^{\lambda_1 t} v_1, e^{\lambda_2 t} v_2]$$

$$\Phi = \begin{bmatrix} e^{-t} & -e^{-6t} \\ 4e^{-t} & e^{-6t} \end{bmatrix}$$

$$\det \Phi = (e^{-t})(e^{-6t}) - (-e^{-6t})(4e^{-t})$$

$$= e^{-7t} + 4e^{-7t} = 5e^{-7t}$$

$$\Phi^{-1} = \frac{1}{5e^{-7t}} \begin{bmatrix} e^{-6t} & e^{-6t} \\ -4e^{-t} & e^{-t} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{5}e^t & \frac{1}{5}e^t \\ -\frac{4}{5}e^{6t} & \frac{1}{5}e^{6t} \end{bmatrix}$$

$$\Phi^{-1} \cdot b = \begin{bmatrix} \frac{1}{5}e^t(6e^{2t}) + \frac{1}{5}e^t(-e^{2t}) \\ -\frac{4}{5}e^{6t}(6e^{2t}) + \frac{1}{5}e^{6t}(-e^{2t}) \end{bmatrix}$$

$$= \begin{bmatrix} \frac{6}{5}e^{3t} - \frac{1}{5}e^{-t} \\ -\frac{24}{5}e^{8t} - \frac{1}{5}e^{4t} \end{bmatrix}$$

$$\int \Phi^{-1} \cdot b = \begin{bmatrix} \frac{18}{5}e^{3t} + \frac{1}{5}e^{-t} \\ -\frac{192}{5}e^{8t} - \frac{4}{5}e^{4t} \end{bmatrix}$$

$$\Phi \cdot \int \Phi^{-1} \cdot b$$

$$= \begin{bmatrix} e^{-t} & -e^{-6t} \\ 4e^{-t} & e^{-6t} \end{bmatrix} \cdot \begin{bmatrix} \frac{18}{5}e^{3t} + \frac{1}{5}e^{-t} \\ -\frac{192}{5}e^{8t} - \frac{4}{5}e^{4t} \end{bmatrix}$$

$$= \begin{bmatrix} e^{-t}(\frac{18}{5}e^{3t} + \frac{1}{5}e^{-t}) - e^{-6t}(-\frac{192}{5}e^{8t} - \frac{4}{5}e^{4t}) \\ 4e^{-t}(\frac{18}{5}e^{3t} + \frac{1}{5}e^{-t}) + e^{-6t}(-\frac{192}{5}e^{8t} - \frac{4}{5}e^{4t}) \end{bmatrix}$$

$$= \begin{bmatrix} \frac{18}{5}e^{2t} + \frac{192}{5}e^{2t} + \frac{1}{5}e^{-2t} + \frac{4}{5}e^{-2t} \\ \frac{72}{5}e^{2t} - \frac{192}{5}e^{2t} + \frac{4}{5}e^{-2t} - \frac{4}{5}e^{-2t} \end{bmatrix}$$

$$x_p = \begin{bmatrix} 42e^{2t} + e^{-2t} \\ -24e^{2t} \end{bmatrix}$$

Check:

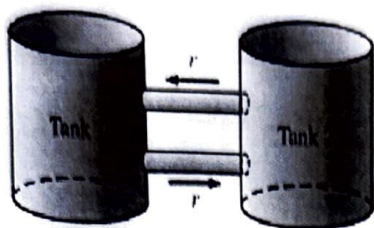
$$x_p' = \begin{bmatrix} 84e^{2t} - 2e^{-2t} \\ -48e^{2t} \end{bmatrix}$$

$$\begin{bmatrix} -5 & 1 \\ 4 & -2 \end{bmatrix} \cdot \begin{bmatrix} x_p \end{bmatrix} = \begin{bmatrix} -5(42e^{2t} + e^{-2t}) + 1(-24e^{2t}) \\ 4(42e^{2t} + e^{-2t}) - 2(-24e^{2t}) \end{bmatrix}$$

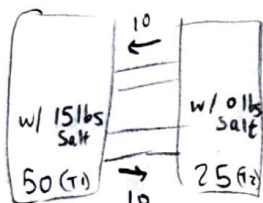
Answer:

$$x_p = \begin{bmatrix} 42e^{2t} + e^{-2t} \\ -24e^{2t} \end{bmatrix}$$

11. The following is a system of two tanks of salt in water:



The volume of Tank 1 is 50 gallons and the volume of Tank 2 is 25 gallons. The rate of flow, r , is 10 gal/min. The initial amount of salt in Tank 1 is 15 pounds and the initial amount of salt in Tank 2 is 0 pounds. Set up and solve the initial value problem using any method. Please show all of your work.



$$X_0 = \begin{bmatrix} 15 \\ 0 \end{bmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} -1/5 - \lambda & 2/5 \\ 1/5 & -2/5 - \lambda \end{vmatrix} = 0$$

$$= \frac{2}{25} + \frac{3}{5}\lambda + \lambda^2 - \frac{2}{25} = 0$$

$$= \lambda(\lambda + 3/5) = 0 \quad \lambda = 0; -3/5$$

$$X'_1 = -10\left(\frac{X_1}{50}\right) + 10\left(\frac{X_2}{25}\right)$$

$$= -1/5 X_1 + 2/5 X_2$$

$$X'_2 = -10\left(\frac{X_2}{25}\right) + 10\left(\frac{X_1}{50}\right)$$

$$= 1/5 X_1 - 2/5 X_2$$

$$X' = \begin{bmatrix} -1/5 & 2/5 \\ 1/5 & -2/5 \end{bmatrix} X$$

$$\lambda = 0$$

$$\begin{bmatrix} -1/5 & 2/5 & 0 \\ 1/5 & -2/5 & 0 \end{bmatrix}$$

$$-1/5 V_1 + 2/5 V_2 = 0$$

$$-1/5 V_1 = -2/5 V_2$$

$$V_1 = 2V_2$$

$$V_2 = V_2$$

$$V_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\lambda = -3/5$$

$$\begin{bmatrix} -1/5 + 3/5 & 2/5 & 0 \\ 1/5 & -2/5 + 3/5 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 2/5 & 2/5 & 0 \\ 1/5 & 1/5 & 0 \end{bmatrix}$$

$$2/5 V_1 = -2/5 V_2$$

$$V_1 = -V_2$$

$$V_2 = V_2$$

$$V_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$X = C_1 e^{0t} \begin{bmatrix} 2 \\ 1 \end{bmatrix} + C_2 e^{-3/5 t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$X(0) = C_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + C_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 15 \\ 0 \end{bmatrix}$$

$$C_2 = 15 - 2C_1$$

$$C_1 = C_2$$

$$C_1 = 5; C_2 = 5$$

$$X = 5 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 5 e^{-3/5 t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$X = \begin{bmatrix} 10 \\ 5 \end{bmatrix} + 5 e^{-3/5 t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

* Check on back

Answer:

$$X = \begin{bmatrix} 10 \\ 5 \end{bmatrix} + 5 e^{-3/5 t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

12. Set up but do not solve the system of differential equations for the following two mass spring system:

$$m_1 = 6; m_2 = 6$$

$$k_1 = 3; k_2 = 2$$

$$m_1 x_1'' = -K_1 x_1 + K_2 (x_2 - x_1)$$

$$m_2 x_2'' = -K_2 (x_2 - x_1) - K_3 x_2$$

$$6x_1'' = -3x_1 + 2(x_2 - x_1)$$

$$= -3x_1 + 2x_2 - 2x_1$$

$$= -5x_1 + 2x_2$$

$$x_1'' = -\frac{5}{6}x_1 + \frac{1}{3}x_2$$

$$6x_2'' = -2(x_2 - x_1)$$

$$= -2x_2 + 2x_1$$

$$= 2x_1 - 2x_2$$

$$x_2'' = \frac{1}{3}x_1 - \frac{1}{3}x_2$$

$$X'' = \begin{bmatrix} -\frac{5}{6} & \frac{1}{3} \\ \frac{1}{3} & -\frac{1}{3} \end{bmatrix} X$$

Answer:

$$X'' = \begin{bmatrix} -\frac{5}{6} & \frac{1}{3} \\ \frac{1}{3} & -\frac{1}{3} \end{bmatrix} X$$

Points: 5 3 0

Math 316 - Exam #5

Name: Seth Recks (2 points will be lost for not writing your name).

Score: 42 / 45 = 93.3 %

1. Find the Laplace transform of $f(t) = e^{-t} \cos(2t)$. You must show all of your work in order to get full credit.

$$\mathcal{L}[\cos 2t] = \frac{s}{s^2 + 4} = F(s)$$

$$a = -1$$

$$F(s-a) = \frac{s+1}{(s+1)^2 + 4}$$

Check:

$$f(s+1) = \frac{s+1}{(s+1)^2 + 4}$$

$$F(s) = \frac{s}{s^2 + 4}$$

$$\mathcal{L}^{-1}[F(s)] = \cos 2t$$

$$\mathcal{L}^{-1}[F(s+1)] = e^{-t} \cos 2t \quad \checkmark$$

Answer:

$$\frac{s+1}{(s+1)^2 + 4}$$

Points: 5 3 0

2. Find the Laplace transform of $f(t) = \frac{1}{2} \sin 6t$. You must show all of your work in order to get full credit.

$$f(t) = \frac{1}{2} \sin 6t$$

$$\mathcal{L}\left[\frac{1}{2} \sin 6t\right] = \frac{1}{2} \mathcal{L}[\sin 6t]$$

$$= \frac{1}{2} \frac{6}{s^2 + 36} = \frac{3}{s^2 + 36}$$

Check:

$$\mathcal{L}^{-1}\left[\frac{3}{s^2 + 36}\right] = \mathcal{L}^{-1}\left[\frac{1}{2} \frac{6}{s^2 + 36}\right]$$

$$f(t) = \frac{1}{2} \sin 6t \quad \checkmark$$

Answer:

$$\frac{3}{s^2 + 36}$$

Points: 5 3 0

3. Find the inverse Laplace transform of $F(s) = \frac{s^2+1}{s^3-2s^2-8s}$ using partial fractions. You must show all of your work in order to get full credit.

$$F(s) = \frac{s^2+1}{s^3-2s^2-8s} = \frac{s^2+1}{s(s^2-2s-8)} = \frac{s^2+1}{s(s-4)(s+2)} = \frac{A}{s} + \frac{B}{s-4} + \frac{C}{s+2}$$

$$s^2+1 = A(s^2-2s-8) + B(s)(s+2) + C(s)(s-4)$$

$$s^2+1 = A(s^2-2s-8) + B(s^2+2s) + C(s^2-4s)$$

$$A+B+C = 1$$

$$-2A+2B-4C = 0$$

$$-8A = 1$$

$$A = -\frac{1}{8}$$

$$-\frac{1}{8} + B + C = 1$$

$$B+C = \frac{9}{8} \quad C = \frac{9}{8} - B$$

$$\frac{2}{8} + 2B - 4C = 0$$

$$2B - 4C = -\frac{2}{8}$$

$$2B - 4\left(\frac{9}{8} - B\right) = -\frac{2}{8}$$

$$2B - \frac{9}{2} + 4B = -\frac{2}{8}$$

$$6B = \frac{17}{4}$$

$$B = \frac{17}{24}$$

$$C = \frac{9}{12}$$

$$f(s) = -\frac{1}{8} \frac{1}{s} + \frac{17}{24} \frac{1}{s-4} + \frac{5}{12} \frac{1}{s+2}$$

$$\mathcal{L}^{-1}[F(s)] = -\frac{1}{8} + \frac{17}{24} e^{4t} + \frac{5}{12} e^{-2t}$$

Check:

$$\mathcal{L}\left[\right] = -\frac{1}{8} \frac{1}{s} + \frac{17}{24} \frac{1}{s-4} + \frac{5}{12} \frac{1}{s+2} = \frac{s^2+1}{s(s-4)(s+2)} = \frac{s^2+1}{s^3-2s^2-8s} \quad \checkmark \checkmark$$

Answer:

$$-\frac{1}{8} + \frac{17}{24} e^{4t} + \frac{5}{12} e^{-2t}$$

Points: 5 3 0

4. Find the inverse Laplace transform for $\tilde{f}(s) = \frac{s^3}{(s-4)^4}$. You must show all of your work in order to get full credit.

$$F(s-4) = \frac{s^3}{(s-4)^4} \quad F(s) = \frac{(s+4)^3}{s^4}$$

$$(s+4)^2 = s^2 + 8s + 16$$

$$(s^2 + 8s + 16)(s+4) = s^3 + 8s^2 + 16s + 4s^2 + 32s + 64$$

$$s^3 + 12s^2 + 48s + 64$$

$$F(s) = \frac{s^3}{s^4} + \frac{12s^2}{s^4} + \frac{48s}{s^4} + \frac{64}{s^4}$$

$$= \frac{1}{s} + \frac{12}{s^2} + \frac{48}{s^3} + \frac{64}{s^4}$$

$$= \frac{1}{s} + 12 \frac{1}{s^2} + 24 \frac{2}{s^3} + \frac{32}{3} \frac{6}{s^4}$$

$$\mathcal{J}^{-1}[F(s)] = 1 + 12t + 24t^2 + \frac{32}{3}t^3$$

$$\mathcal{J}^{-1}[F(s-4)] = e^{4t} \left[1 + 12t + 24t^2 + \frac{32}{3}t^3 \right]$$

Check:

$$\mathcal{J}\left[\frac{1}{s-4}\right] = \frac{1}{s-4} + \frac{12}{(s-4)^2} + 24 \frac{2}{(s-4)^3} + \frac{32}{3} \frac{6}{(s-4)^4} = \frac{s^3}{(s-4)^4} \quad \checkmark \checkmark$$

Answer:

$$e^{4t} \left[1 + 12t + 24t^2 + \frac{32}{3}t^3 \right]$$

Points: 5 3 0

5. Solve the initial value problem $y'' + 4y = \sin 3t$, $y(0) = 0$, $y'(0) = 0$ using Laplace transforms. You must show all of your work in order to get full credit.

$$\mathcal{L}[y''] + 4\mathcal{L}[y] = \mathcal{L}[\sin 3t]$$

$$s^2 F(s) - (s)(0) - 0 + 4F(s) = \frac{3}{s^2 + 9}$$

$$(s^2 + 4)F(s) = \frac{3}{s^2 + 9}$$

$$F(s) = \frac{3}{(s^2 + 9)(s^2 + 4)} = \frac{As + B}{s^2 + 9} + \frac{Cs + D}{s^2 + 4} = -\frac{3}{5} \frac{1}{s^2 + 9} + \frac{3}{5} \frac{1}{s^2 + 4}$$

$$3 = (As + B)(s^2 + 4) + (Cs + D)(s^2 + 9)$$

$$= As^3 + Bs^2 + 4As + 4B$$

$$+ Cs^3 + Ds^2 + 9Cs + 9D$$

$$\mathcal{L}^{-1}\left[-\frac{1}{5} \frac{3}{s^2 + 9}\right] = -\frac{1}{5} \sin 3t$$

$$\mathcal{L}^{-1}\left[\frac{3}{10} \frac{2}{s^2 + 4}\right] = \frac{3}{10} \sin 2t$$

$$\mathcal{L}^{-1}[F(s)] = -\frac{1}{5} \sin 3t + \frac{3}{10} \sin 2t = y$$

Check:

$$y = -\frac{1}{5} \sin 3t + \frac{3}{10} \sin 2t$$

$$y' = -\frac{3}{5} \cos 3t + \frac{3}{5} \cos 2t$$

$$y'' = \frac{9}{5} \sin 3t - \frac{6}{5} \sin 2t$$

$$y'' + 4y = \sin 3t$$

$$\frac{9}{5} \sin 3t - \frac{6}{5} \sin 2t - \frac{4}{5} \sin 3t + \frac{6}{5} \sin 2t = \sin 3t$$

$$\sin 3t = \sin 3t \checkmark$$

Answer:

$$y = -\frac{1}{5} \sin 3t + \frac{3}{10} \sin 2t$$

Points: 5 3 0

6. Solve the initial value problem $y'' + 3y' + 2y = t$, $y(0) = 0$, $y'(0) = 2$ using Laplace transforms. You must show all of your work in order to get full credit.

$$\mathcal{L}[y''] + 3\mathcal{L}[y'] + 2\mathcal{L}[y] = \mathcal{L}[t]$$

$$s^2 F(s) - (s)(0) - 2 + 3[sF(s) - 0] + 2F(s) = \frac{1}{s^2}$$

$$s^2 F(s) - 2 + 3sF(s) + 2F(s) = \frac{1}{s^2}$$

$$s^2 F(s) + 3sF(s) + 2F(s) = \frac{1}{s^2} + 2$$

$$F(s)(s^2 + 3s + 2) = \frac{1}{s^2} + 2$$

$$F(s) = \frac{1}{s^2(s^2 + 3s + 2)} + \frac{2}{s^2 + 3s + 2}$$

$$= \frac{1}{s^2(s+1)(s+2)} + \frac{2s^2}{(s+1)(s+2)s^2} = \frac{2s^2 + 1}{(s^2)(s+1)(s+2)}$$

$$= \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+1} + \frac{D}{s+2}$$

$$2s^2 + 1 = A(s)(s^2 + 3s + 2) + B(s^2 + 3s + 2) + C(s^2)(s+2) + D(s+1)(s^2)$$

$$= A(s^3 + 3s^2 + 2s) + B(s^2 + 3s + 2) + C(s^3 + 2s^2) + D(s^3 + s^2)$$

$$A + C + D = 0$$

$$3A + B + 2C + D = 2$$

$$2A + 3B = 0$$

$$2B = 1$$

$$B = \frac{1}{2}$$

$$2A + \frac{3}{2} = 0$$

$$2A = -\frac{3}{2}$$

$$A = -\frac{3}{4}$$

$$-\frac{3}{4} + C + D = 0$$

$$D = \frac{3}{4} - C$$

$$3(-\frac{3}{4}) + \frac{1}{2} + 2C + \frac{3}{4} - C = 2$$

$$C - 1 = 2$$

$$C = 3$$

$$D = -\frac{9}{4}$$

$$F(s) = -\frac{3}{4} \frac{1}{s} + \frac{1}{2} \frac{1}{s^2} + 3 \frac{1}{s+1} - \frac{9}{4} \frac{1}{s+2}$$

$$\mathcal{L}^{-1}[F(s)] = -\frac{3}{4} + \frac{1}{2}t + 3e^{-t} - \frac{9}{4}e^{-2t} = y$$

* Check on back

Answer:

$$y = -\frac{3}{4} + \frac{1}{2}t + 3e^{-t} - \frac{9}{4}e^{-2t}$$

Points: 5 3 0

Check:

$$y'' + 3y' + 2y = t$$

$$y = -\frac{3}{4} + \frac{1}{2}t + 3e^{-t} - \frac{9}{4}e^{-2t}$$

$$y' = \frac{1}{2} - 3e^{-t} + \frac{9}{2}e^{-2t}$$

$$y'' = 3e^{-t} - 9e^{-2t}$$

$$3e^{-t} - 9e^{-2t}$$

$$+ 3\left[\frac{1}{2} - 3e^{-t} + \frac{9}{2}e^{-2t}\right]$$

$$+ 2\left[-\frac{3}{4} + \frac{1}{2}t + 3e^{-t} - \frac{9}{4}e^{-2t}\right]$$

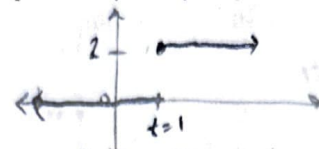
$$\cancel{3e^{-t}} - \cancel{9e^{-2t}}$$

$$+ \cancel{3/2} - \cancel{9e^{-t}} + \cancel{27/2}e^{-2t}$$

$$-\cancel{3/2} + t + \cancel{6e^{-t}} - \cancel{9/2}e^{-2t}$$

$$t = t \quad \checkmark \checkmark$$

7. Solve the initial value problem $y'' + y = f(t)$, $y(0) = 0$, $y'(0) = 0$ where

$$f(t) = \begin{cases} 0 & t < 1 \\ 2 & t \geq 1 \end{cases}$$


$f(t) = 2u(t-1)$ at $t=1$
 ↑ 2 gets "turned on"

You must use Laplace transforms and must show ALL of your work.

$$y'' + y = 2u(t-1), \quad y(0) = 0, \quad y'(0) = 0$$

$$\mathcal{L}[y''] + \mathcal{L}[y] = \mathcal{L}[2u(t-1)] \quad f(t) = 1$$

$$s^2 F(s) + F(s) = 2 \cdot e^{-s} \cdot \frac{1}{s} = \frac{2e^{-s}}{s}$$

$$F(s)(s^2 + 1) = \frac{2e^{-s}}{s}$$

$$F(s) = \frac{2e^{-s}}{s(s^2 + 1)}$$

$$\frac{2}{s(s^2 + 1)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 1} = \frac{2}{s} - \frac{2s}{s^2 + 1}$$

$$2 = A(s^2 + 1) + (Bs + C)s$$

$$= A(s^2 + 1) + Bs^2 + Cs$$

$$A + B = 0$$

$$C = 0$$

$$A = 2$$

$$B = -2$$

$$\mathcal{L}^{-1}\left[\frac{2}{s} - \frac{2s}{s^2 + 1}\right] = 2 - 2\cos t$$

$$\mathcal{L}^{-1}[F(s)] = u(t-1)[2 - 2\cos t]$$

$$y = 2u(t-1)[1 - \cos t]$$

Check not possible :
 Don't know derivatives of
 unit step functions.

Answer:

$$y = 2u(t-1)[1 - \cos(t-1)]$$

Points: 5 (3) 0

(4)

8. Solve the initial value problem $y'' + 4y' + 13y = \delta(t-3)$, $y(0) = 1$, $y'(0) = 0$. You must use Laplace transforms and show all of your work in order to get full credit.

$$\mathcal{L}[y''] + 4\mathcal{L}[y'] + 13\mathcal{L}[y] = \mathcal{L}[\delta(t-3)]$$

$$s^2 F(s) - (s)(1) - 0 + 4[sF(s) - 1] + 13F(s) = e^{-3s}$$

$$s^2 F(s) - s + 4sF(s) - 4 + 13F(s) = e^{-3s}$$

$$s^2 F(s) + 4sF(s) + 13F(s) = e^{-3s} + s + 4$$

$$F(s) = \frac{e^{-3s}}{s^2 + 4s + 13} + \frac{s+4}{s^2 + 4s + 13}$$

$$\frac{s+4}{s^2 + 4s + 13} = \frac{s+4}{s^2 + 4s + 4 + (13-4)} = \frac{s+4}{(s+2)^2 + 9} = F(s+2)^{\text{New}}$$

$$F(s) = \frac{s-2+4}{s^2 + 9} = \frac{s+2}{s^2 + 9} = \frac{s}{s^2 + 9} + \frac{2}{s^2 + 9} = \frac{s}{s^2 + 9} + \frac{2}{3} \frac{3}{s^2 + 9}$$

$$\mathcal{L}^{-1}[F(s)] = \cos 3t + \frac{2}{3} \sin 3t$$

$$\mathcal{L}^{-1}[F(s+2)] = e^{-2t} \left[\cos 3t + \frac{2}{3} \sin 3t \right]$$

$$\frac{1}{s^2 + 4s + 13} = \frac{1}{s^2 + 4s + 4 + (13-4)} = \frac{1}{(s+2)^2 + 9} = F(s+2)$$

$$F(s) = \frac{1}{s^2 + 9} \quad \mathcal{L}^{-1}[F(s)] = \frac{1}{3} \sin 3t$$

$$\mathcal{L}^{-1}[F(s+2)] = \frac{1}{3} e^{-2t} \sin 3t$$

$$\rightarrow \mathcal{L}^{-1}[F(s)] = u(t-3) \left[\frac{1}{3} e^{-2(t-3)} \sin(3(t-3)) \right] + e^{-2t} \left[\cos 3t + \frac{2}{3} \sin 3t \right]$$

$$y = \frac{1}{3} u(t-3) e^{-2t+6} \sin(3t-9) + e^{-2t} \left[\cos 3t + \frac{2}{3} \sin 3t \right]$$

Good luck checking
that answer!

Answer:

$$y = \frac{1}{3} u(t-3) e^{-2t+6} \sin(3t-9) + e^{-2t} \left[\cos 3t + \frac{2}{3} \sin 3t \right]$$

Points: 5 (3) 0

9. Solve the initial value problem $y' + 2y = 0, y(0) = 1$. You must use Laplace transforms and show all of your work in order to get full credit.

$$\mathcal{L}[y'] + 2\mathcal{L}[y] = \mathcal{L}[0]$$

$$sF(s) - 1 + 2F(s) = 0$$

$$sF(s) + 2F(s) = 1$$

$$F(s) = \frac{1}{s+2}$$

$$\mathcal{L}^{-1}[F(s)] = e^{-2t} = y$$

Check:

$$y' = -2e^{-2t}$$

$$y' + 2y = -2e^{-2t} + 2e^{-2t} = 0 \checkmark \checkmark$$

Answer:

$$y = e^{-2t}$$

Points: 5 3 0

Laplace Transforms

Laplace of Basic Functions: $\mathcal{L}[f(t)] = F(s)$

- $\mathcal{L}[e^{at}] = \frac{1}{s-a}$
- $\mathcal{L}[t^n] = \frac{n!}{s^{n+1}}$
- $\mathcal{L}[\sin(at)] = \frac{a}{s^2+a^2}$
- $\mathcal{L}[\cos(at)] = \frac{s}{s^2+a^2}$
- $\mathcal{L}[\sinh(at)] = \frac{a}{s^2-a^2}$
- $\mathcal{L}[\cosh(at)] = \frac{s}{s^2-a^2}$

Laplace Rules

- $\mathcal{L}[c \cdot f(t)] = c \cdot \mathcal{L}[f(t)] = cF(s)$
- $\mathcal{L}[f(t) + g(t)] = \mathcal{L}[f(t)] + \mathcal{L}[g(t)] = F(s) + G(s)$
- $\mathcal{L}[e^{at} \cdot f(t)] = F(s-a)$ – Find the Laplace of $f(t)$ then replace s with $s-a$
- $\mathcal{L}[t^n \cdot f(t)] = (-1)^n F^{(n)}(s)$ – Find Laplace of $f(t)$; find the n th derivative of this; multiply by this by $(-1)^n$

Laplace of Heaviside and Dirac Delta Functions

- $\mathcal{L}[u(t-c) \cdot f(t)] = e^{-cs} \cdot \mathcal{L}[f(t+c)]$ (translate before transform)
- $\mathcal{L}[u(t-c) \cdot f(t-c)] = e^{-cs} \mathcal{L}[f(t)] = e^{-cs} F(s)$
- $\mathcal{L}[\delta(t-c)] = e^{-cs}$

Laplace of Derivatives

- $\mathcal{L}[f'(t)] = sF(s) - f(0)$
- $\mathcal{L}[f''(t)] = s^2 F(s) - sf(0) - f'(0)$